

Rotterdam School of Management - Erasmus University

Master Thesis – M.Sc. in Finance & Investments

How does economic policy uncertainty affect the cross-border M&A activity & the method of payment in cross-border M&A deals?

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Author: Alexandros Dimitriadis

Student Number: 590887ad

Coach: Dr. Farooq Ahmad

Co-reader: Dr. Arjen Mulder

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Abstract

The purpose of this thesis is to examine how economic policy uncertainty (EPU), measured by the economic policy uncertainty index developed by Baker, Bloom & Davis (BBD index), affects the cross-border M&A activity and the method of payment in cross-border M&A deals, by using a country sample of 24 countries and a sample of announced deals over the period between 2010 and 2019. Firstly, a significant relationship between the EPU and the outbound M&A activity could not be established; however, the results show a significant negative relationship between EPU and the inbound M&A activity, whilst this effect becomes more prominent when a country is affected by high EPU levels for a longer period. Secondly, no significant relationship could be found between the effect of EPU on cross-border M&A activity and the institutional quality of a country. Lastly, the effect of the target country's EPU on the bidders' preferences for stock payments was found to be significantly positive, while the same effect of the acquirer country's EPU was found to be positive, but not statistically significant. These results are consistent with the hypothesis that bidders are reluctant to exchange their liquid assets with the illiquid assets of the target in cross-border transactions, when they face high EPU levels.

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1. Introduction

Mergers and acquisitions (hereafter mentioned as M&As, deals or transactions) represent a very useful tool for companies and CEOs who wish to expand their operations. Moreover, they can also be a form of investment for companies and investment vehicles, such as private equity funds, which have limited strategic goals and are primarily concerned about the returns that the subsequent sale of the acquired companies can provide. By utilizing various strategies, ranging from creating synergies, increasing market shares and enforcing efficient structural and operational changes, to simply replacing underperforming managers, acquirers aim at increasing the value of the stakeholders. Consequently, the outcome of such transactions is of the utmost importance, not only for the companies themselves, but also for all other constituents in them (Sudarsanam, 2010).

Despite the temporary deceleration caused by the 2020 covid-19 global pandemic, both the number and the volume of deals in the seventh M&A wave continued to rise in 2021. Specifically, the volume of global M&A deals reached an all-time high of c.\$5.9 trillion in 2021 (Thomson Reuters). Cross-border M&A deals have also become increasingly more popular, facilitated by the increased globalization efforts, which have led to greater flows of capital, goods and services across borders (Norbäck & Persson, 2008). Between 2014 and 2020, the volume and number of international M&A deals increased significantly, with the volume reaching an all-time high of c.\$2.1 trillion in 2021, showing a 69% increase from the 2020 figure, while the number of cross-border deals amounted to 8,517 in 2021 (Thomson Reuters). These record numbers illustrate the importance companies and CEOs place on cross-border deals as a type of corporate investment.

It is a general and logical assumption for cross-border deals to be perceived as riskier and as having a higher level of asymmetric information affecting them, compared to domestic deals, due to the existence of and the distance between national borders. Events such as the Brexit, tense national elections and frictions between neighboring nations can create significant uncertainty and instability. In turn, uncertainty can lead to actual implications affecting the behavior of economic agents (Bloom et al., 2007; Bloom, 2009). As per Brogaard & Detzel (2015), governmental unreliability can further contribute to uncertainty regarding fiscal, regulatory and monetary policy. The uncertainty relating to economic and fiscal policies and decisions is described as economic policy uncertainty (hereafter mentioned as EPU). EPU can be caused by various reasons including natural disasters, geopolitical shifts and changes in the underlying macro-economic cornerstones

of a country (Baker et al., 2016). The 2008 global financial crisis raised the awareness of the public regarding the importance of EPU, as this crisis brought forward a number of oversights in regulations. Additionally, EPU has increased in the last decade, due to events such as the Brexit, hostile attitudes between countries, such as the Ukrainian crisis and environmental problems threatening the globe. During the period between 2008 and 2018, the average EPU was two times higher than that of the periods between 1988 and 1998 and 1998 and 2008 (Baker et al., 2016).

Besides that, the interest in EPU amongst academics rose due to a new measurement method developed by Baker, Bloom & Davis (2016). The Baker, Bloom & Davis index (hereafter mentioned as the BBD index or simply EPU) is a proxy for the EPU, measuring the frequency of components or terms referring to economy, policy and uncertainty in newspaper articles within a country. The BBD index is considered a more comprehensive measure of a nation's economic policy uncertainty amongst researchers, relative to other proxies such as national elections and proxies based on investors' sentiments. Specifically in the field of M&As, researchers have already engaged with EPU and how it affects cross-border deals. However, research on this topic is rather limited and contradicting. Certain scholars focus on the relation between EPU and domestic M&A activity (Bonaime et al., 2018; Borthwick et al., 2020; Nguyen & Phan, 2017; Sha et al., 2020), while others investigate the effect of EPU on the likelihood of cross-border M&As (Cao et al., 2017; Lee, 2018; Li et al., 2021). Using the BBD index, Nguyen & Phan (2017) and Bonaime et al. (2018) suggest that high uncertainty is negatively correlated with the M&A activity in the U.S. market. However, Sha et al. (2020) have found that companies are more likely to engage in M&As in China when economic uncertainty is high. This level of contradiction in the outcomes of various scholars dictates that there is still space for further research on the topic.

An important consideration in M&A deals is the method of payment. The choice and availability of different methods of payment can play a decisive role in the outcome of the whole deal, as, for instance, in cases where the seller does not accept payment by stock, because the seller might believe that the bidder's stock is overvalued, bidders without available cash cannot proceed to conclude the deal. The choice of the means of payment, mainly between cash or stock, is governed by various factors, such as information asymmetry problems, firm-specific characteristics, for instance the firm size and the cash holdings of the acquirer, and country-specific characteristics, for example tax regulations, political uncertainty and of course EPU (Franks et al., 1998; Shleifer & Vishny, 2003; Alexandridis et al., 2012; Boone et al., 2014; Nguyen & Phan, 2017). Previous

literature suggests that a higher level of information asymmetry is associated with cross-border deals, than that associated with domestic deals, leading bidders to use stock as the main payment method (Nguyen & Phan, 2017), as they have less incentives to exchange liquid cash for other illiquid assets of the target, i.e. the target's assets or shares.

Due to the increased interest in EPU and in the cross-border M&A activity and the contradicting outcomes in previous research, this thesis aims at investigating the inbound and outbound cross-border M&A deal flow and the bidders' preferences for stock payments in relation to the EPU. The specific research question to support these goals can be formulated as follows:

How does economic policy uncertainty affect the cross-border M&A activity & the method of payment in cross-border M&A deals?

The methodologies which will be used to explore the abovementioned question are similar to those utilized by Cao et al. (2017) and Nguyen & Phan (2017). However, by using the BBD index as a proxy for EPU, and by exploring cross-border deals in relation to the method of payment, this thesis differentiates itself from previous scholars who have used other measurements of EPU, such as the study by Cao et al. (2017) which utilized national elections as the proxy for EPU. This study employs a country sample consisting of 24 countries, while the M&A sample includes cross-border M&A deals with an announcement date between 2010 and 2019.

This thesis provides evidence that EPU inhibits the inbound M&A activity, following Cao et al. (2017), and supporting the delaying hypothesis as the underlying driver of this relationship. However, the relationship between EPU and the outbound activity was found negative, but not statistically significant, contradicting the results of Cao et al. (2017), and the hedging hypothesis as the underlying driver of the relationship. As in the case of the relationship between EPU and the inbound activity, the delaying hypothesis seems to be the driver behind the effect of EPU on the outbound activity. Additionally, the hypothesis that the effect of EPU on the cross-border M&A activity is smaller in countries with higher levels of institutional quality could not be empirically supported, mainly due to measurement issues regarding the quality of institutions variable utilized in this study. Lastly, according to the information asymmetry theory and similar to the results of Nguyen & Phan (2017), the effect of EPU on the bidders' preferences for stock payments was found significantly positive, supporting the idea that bidders are reluctant to pay with liquid assets, e.g. cash, for the target's illiquid assets, e.g. the target's stock, when they face high EPU levels.

2. Literature review

An event such as a M&A is one of the most important decisions and processes in the life of a company. A merger is the combination of two, usually similar, distinct entities into a single legal entity, while an acquisition is the takeover of one company by another. M&As can be categorized based on their strategic goals. The first type of strategic M&As is vertical mergers, which happen when one company purchases another that falls in a different place along the same supply chain. The second type is horizontal acquisitions, which refer to deals where companies buy other firms in the same industry, i.e. competitors. The aim of such a deal is not to merge two different stages of the same supply chain, as in the first category, but to combine two competing players, increasing the acquirer's market share. The third type is market extension acquisitions. Although very similar to horizontal mergers, the two companies are not competitors in this case, as they operate in different markets. For example, a market extension acquisition occurs when one company buys another, in the same industry, but in another country. Lastly, a conglomerate acquisition occurs between companies in different industries. This categorization of M&A deals can be useful in order to understand why the interest in and the volume of (cross-border) M&A activity have increased after the 2008 crisis, facilitated by the increasing globalization efforts.

2.1. The seventh wave of M&A activity

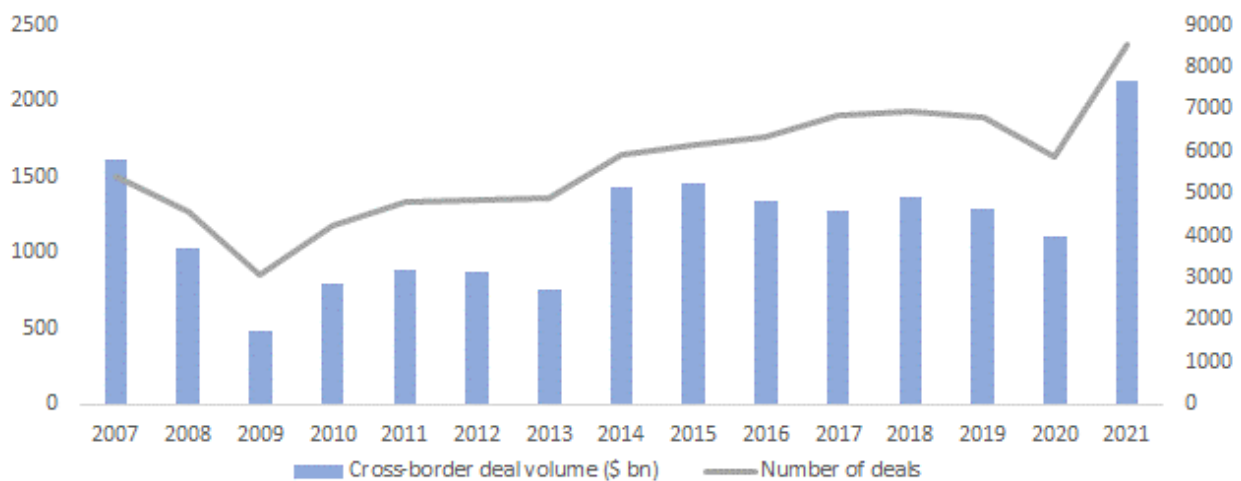
Mergers and acquisitions are becoming an increasingly more popular form of corporate development, as companies and management teams strive to increase their value and their power. The global market is currently going through its seventh wave of M&A activity. This latest wave was initiated after the 2008 U.S. mortgage bubble crisis, which in turn triggered the 2008 global financial crisis. In 2007, i.e. the last year of the sixth wave of M&A activity, global M&A volume recorded, at that time, an all-time high of c.\$4.55 trillion (Thomson Reuters). Despite the downturn in the years following the 2008 crisis, M&As remained a very popular expansion tool for companies, as global M&A volume amounted to c.\$4.2 trillion in 2015 and, following a small deceleration, to c.\$3.8 trillion in 2019 (Thomson Reuters). The initial market turmoil caused by the global covid-19 pandemic in 2020 resulted in a slight decrease, of c.6%, in the deal volume in 2020, as the uncertainty about the pandemic led many companies to postpone their deals in that year. The pandemic also led to increased traffic in online retailers and to increased usage of software products. Thus, in 2021, the projects postponed in 2020 and the increased interest in IT, software and online retailing companies, drove the global M&A volume to a new all-time high of c.\$5.9

trillion (Thomson Reuters). These record numbers highlight how important M&As have become as a corporate practice for expansion, incentivizing researchers to explore the various factors affecting them.

2.2. Cross-border M&A activity during the seventh wave of M&As

Following the increased volume of global M&A activity, the volume of and the interest in cross-border M&As also increased during the seventh wave of M&A activity. As noted previously, the new age of globalization presents numerous opportunities for firms to enter and profit from new, unexplored markets, acquire patents and proprietary technology from technological hubs in eastern countries and overall expand their operations across national borders. The aggregate dollar volume of cross-border deals increased from 21% of the worldwide acquisition activity to 37%, in the period between 1996 and 2014 (Cao, Li, & Liu, 2017). Between 2014 and 2020, the number and volume of international M&A deals continued to increase, and despite the problems the 2020 covid-19 pandemic created in the global market, the total volume of cross-border deals in 2021 reached an all-time high of c.\$2.1 trillion, showing a 69% increase from the 2020 figure, while the number of cross-border deals amounted to 8,517 in 2021 (Thomson Reuters). This increased volume of cross-border M&As has led researchers to explore the underlying factors which influence international M&A activity. The following graph presents further information about the historical volume and number of cross-border M&A deals.

Graph 2.2.1.: Global cross-border M&A activity



2.3. Factors affecting cross-border M&As

The existence of national borders poses a further item for consideration for companies when screening for international M&A targets, compared to domestic deals. Several previous studies have extensively examined numerous determinants of cross-border M&A activity. Scholars have already identified significant connections between the macroeconomic and financial market environment and the cross-border M&A activity, which are usually approximated by the gross domestic product (GDP), the GDP per capita, the stock market capitalization, the trade activity between country-pairs and the exchange and inflation rates (Erel et al., 2012; Hyun & Kim, 2010; Rossi & Volpin, 2004; Uddin & Boateng, 2011).

Further research examines the relationship between M&A transactions and the geographical and cultural differences between the target and acquirer countries (Ahern, Daminelli, & Fracassi, 2015; Morosini, Shane, & Singh, 1998), while Guardo et al. (2016) investigate the effect the political environment and corruption can have on M&A activity. Moreover, Huizinga & Voget (2009) look into the role of taxation and tax-related regulation in M&A decisions and activities, while Ferreira et al. (2010) delve into the connection between foreign institutional ownership and cross-border M&As. Existing literature also explores the link between the cross-border M&A likelihood and the target, or host, country's institutional and regulatory systems, case in point being the accounting standards and investor protection rights (Rossi & Volpin, 2004), the creditor rights (Renneboog et al., 2017) and the employment protection (Dessaint et al., 2017).

Additionally, previous literature has provided evidence for the positive relationship between M&A activity and the quality of institutions of a country. Peng et al. (2008) and Dinc & Erel (2013) argue that institutions play a greater role for emerging market acquirers due to government involvement. Additionally, Dikova et al. (2010) examined the influence of deal regulatory bodies and the quality of institutional environments on the deal duration. This focus on the quality of institutions suggests that it can be an influential factor to the cross-border M&A process, however as far as known, research specifically connecting EPU to the quality of institutions of a country is rather limited.

Overall, this vast body of literature has identified numerous, mainly macroeconomic, potentially influential to the cross-border M&A process factors and has provided evidence for their importance. The extensive research on the topic also indicates the increased importance both academics and practitioners place on the cross-border M&A process. Despite, however, the

numerous and diverse fields of the topic which have already been explored, the literature engaged with uncertainty, specifically uncertainty related to economic and governmental policy, and the effects it can have on cross-border M&As, is rather limited.

2.3.1. Uncertainty & cross-border M&As

It is a general belief that uncertainty could lead to decreased investment activity, as the delaying hypothesis states that investors would postpone their actions until a certain level of confidence about both country and firm-specific prospects can be secured. This principle should also apply to corporate M&A investments. First of all, a distinction must be made between two types of cross-border M&A deals. For a country of interest, these types are the outbound and the inbound deals. Outbound deals are M&As where a company within a specified country, hereafter also called the home or the acquirer country of the outbound deal, acquires a company based in another country, also mentioned as the host or target country. On the other hand, inbound M&As are deals where a company based outside of the country of interest acquires a company based within it.

Uncertainty in a country could deter outside investors from engaging in cross-border deals and thus lead to decreased inbound deal volumes. For instance, Dinc & Erel (2013) argue that doubts about nationalism during elections hinder inbound M&A activity, as nationalistic government actions typically are against the interest of foreign investors. In line with the real option theory, as uncertainty in a country increases, the value of the real option of the investment also increases, thus foreign investors have incentives to postpone their cross-border investments targeting that country.

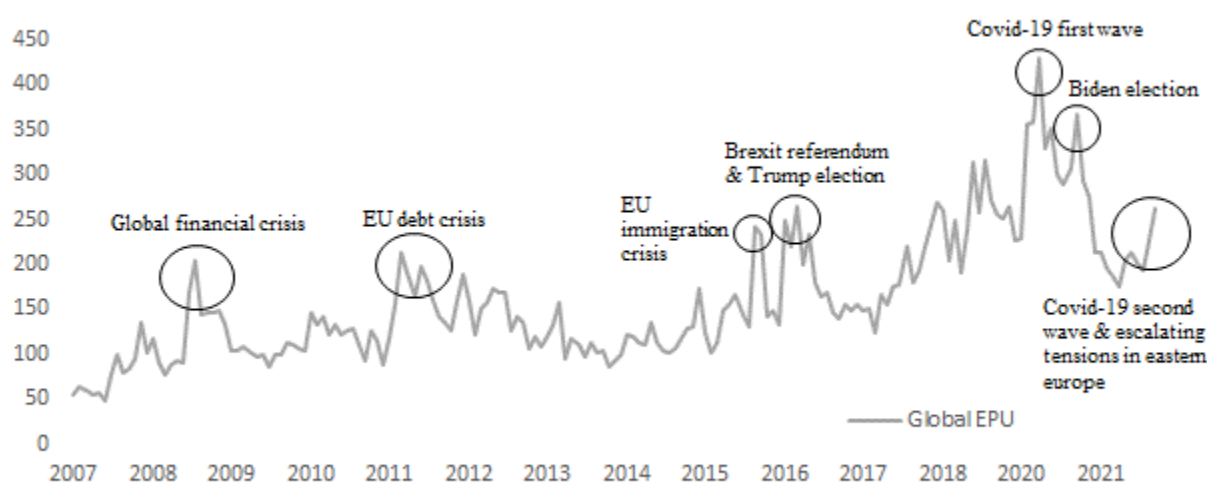
Prior research provides evidence that increased market implied volatility, which could be a proxy for market uncertainty, results in reduced M&A activity (Bhagwat et al., 2016). Garfinkel & Hankins (2011), by utilizing the past volatility of the operating income and the volatility of the cost of goods sold as proxies for the cash flow uncertainty, showed that higher future cash flow uncertainty increases the likelihood for vertical mergers. Cao et al. (2017) identified a negative connection between political uncertainty, approximated by national elections, and inbound M&A deals. Specifically, they have found that upcoming elections lead to increased outbound (based on the hedging hypothesis) and decreased inbound (based on the delaying hypothesis) cross-border M&A activity. As a special type of uncertainty, EPU is expected to have similar effects on cross-border M&A activity.

2.3.2. Economic policy uncertainty

Galbraith (1977) introduced uncertainty as a macroeconomic factor potentially affecting corporate decisions, prompting researchers to further explore various types of uncertainty in relation to corporate decisions. Al-Thaqeb & Algharabali (2019) defined economic uncertainty as unexpected changes that influence the economic ecosystem. Geopolitical uncertainty, regulatory uncertainty and economic policy uncertainty are a few cases of economic uncertainty. EPU is the type of uncertainty which arises when there are serious doubts about what economic policies are about to be taken, who will make these decisions, when will these policies take place and what will their effects be (Baker et al., 2006). EPU can be created by several causes and events such as natural disasters, geopolitical shifts and changes in macro-economic factors (Baker et al., 2016), as mentioned earlier.

Baker, Bloom & Davis (2016) created the BBD index, which quantifies the level of EPU based on the newspaper coverage of policy-related economic uncertainty. In close coordination with the original creators of the BBD index, other researchers created indices in a similar manner for a number of countries worldwide. As a more comprehensive measure of EPU, the creation of the BBD index facilitated research in the field of EPU and sparked the interest of scholars. Additionally, the increased academic interest in EPU was also facilitated by the increased level of uncertainty during the past years, starting from the 2008 global crisis. The graph below illustrates this increase in global EPU, as approximated by the BBD index, in relation to certain triggering events.

Graph 2.3.2.: Historical global EPU (BBD index)



Source: Baker, Davis & Bloom website (economic policy uncertainty website).

These high levels of uncertainty have also led government officials to redefine their focus on and their methods of dealing with EPU, as more and more events caused turmoil during the last decade.

In contrast with other types of uncertainty, EPU can be managed to a certain extent by policy-makers, i.e. governments, central banks and other regulatory bodies. Various types of government involvement in the market, such as subsidies and changes in tax and law regulations, can significantly affect the private business sector (Pástor & Veronesi, 2012), and consequently the likelihood and volume of M&As. Hence, regulatory bodies and government officials can use EPU to indirectly influence corporate investments and M&A activities in their countries. Therefore, research on EPU could have important practical implications, both for regulatory bodies and for companies.

2.4. Method of payment & economic policy uncertainty

The second part of this thesis will focus on how EPU is related to the method of payment in cross-border M&A deals. The method of payment is an important consideration in a transaction, as the willingness of the seller to accept different payment methods, and the availability of these methods to the bidder, can significantly influence the outcome of a deal (Nguyen & Phan, 2017). The majority of M&A deals are financed with either cash or equity, or a mixture of both, with some deals even including other types of assets, such as bonds and specific types of receivables. Historically, cash has been the most popular method of payment, as cash payments are associated with positive market-signaling effects (Shleifer & Vishny, 2003; Alexandridis et al., 2012; Boone et al., 2014). Stock payments, on the other hand, have become increasingly more common, especially during periods of strong market performance (Shleifer & Vishny, 2003; Alexandridis et al., 2012; Boone et al., 2014), as a measure of mitigating the risk of overpayment.

Previous literature have already explored different factors affecting the choice of the payment method in cross-border deals. For instance, tax regulations and tax incentives directly affect the choice of the payment method (Franks, Harris & Mayer, 1988). Additionally, researchers, including Wansley et al. (1987) and Yook et al. (1999), argue that the payment method choices in acquisition deals are driven by asymmetric information, with acquirers preferring non-cash, i.e. stock, payments when there is considerable valuation uncertainty. Specifically, buyers try to mitigate the risk of overpayment by using stock. This is especially true when the management of the acquiring firm has inside information that their firm's stock is overvalued (Myers & Majluf, 1984). Besides

that, information asymmetry problems in cross-border transactions can also arise from country-specific factors, such as EPU.

Empirical studies have reached to the conclusion that cross-border M&As generally involve higher information asymmetry levels compared to domestic transactions, as mentioned before. Scholars have also proposed further potential reasons for this phenomenon, including the inequality in shareholder protection (Erel et al., 2012) and the regulatory and geographical distances between the home and host countries (Rossi & Volpin, 2004). According to Erel et al. (2012) and Huang et al. (2016), these country-specific characteristics have an effect on the method of payment of acquisitions. Specifically, Huang et al. (2016) suggest that the acquirer could face less governance risk, related to the target's country, when paying with stock. Additionally, in periods of high policy uncertainty, target companies might be more vulnerable to financial constraints and default risks, as future cash flows become riskier under periods of high policy uncertainty (Nguyen & Phan, 2017). This results to fewer incentives for the acquirers to exchange liquid cash for less liquid assets, i.e. the target's shares and assets. Using the same underlying reasoning, this thesis will investigate how EPU can affect the choice of the payment method in cross-border M&A deals.

In order to put things into perspective, a real world case should be examined. In 2016, Enbridge Inc., a Canadian energy infrastructure company, announced its intentions to acquire and merge with Spectra Energy Corporation, a U.S. based company, a deal which was completed in 2017. As shown in graph 2.3.2., EPU rose sharply during the period leading to and after the Trump election, especially in the U.S., as this is the country that would be affected the most by the election's outcome. As the target company in the specific deal is based in the U.S., and the ultimate goal of the transaction was to combine the operations of the two entities, the aforementioned deal was fully paid for in stock, as the acquirer's plan to merge and create the largest energy infrastructure company in North America could be affected significantly by the potential instability in the region, created after the Trump election. Moreover, at the time of the deal, the acquirer's management team most probably believed that the election's results would hinder the stock performance of their company, thus they preferred to fund the deal with their own stock, which they believed would later lose value, tying the decision back to the information asymmetry problem described above.

2.5. Hypotheses development

Building upon previously discussed theories, this section of the thesis provides an overview of the hypotheses under question.

The hedging hypothesis suggests that companies may choose cross-border M&As as a way to diversify risk arising from policy uncertainty, as global diversification could enhance operating flexibility within the firm and help the firm respond to political, currency, market risks and other institutional differences (Berger, 2000; Denis, Denis & Yost, 2002). Furthermore, geographical diversification can satisfy investors' diversification preferences (Brewer, 1981; Fatemi, 1984). Scholars have shown that home country political risk factors encourage firms to perform foreign investments (Tallman, 1988; Le & Zak, 2006). Hence, the first hypothesis, similar to the hypothesis tested by Cao et al. (2017), will explore the connection between EPU and the outbound M&A activity.

Hypothesis 1: A higher EPU index is associated with increased cross-border M&A outflow.

The delaying hypothesis proposes that higher uncertainty causes firms to become cautious and postpone or even abort their investments. According to Bernanke (1983) and Bloom et al. (2007), corporate investment projects are costly or even impossible to reverse. This leads companies to consider their actions more carefully and wait to act, in order to avoid costly mistakes. Hence, it is only natural to assume that higher EPU levels should lead to decreased inbound M&A activity. Thus, based on the delaying hypothesis, the second hypothesis of this thesis, similar to the one tested by Cao et al. (2017), can be formulated as follows, in order to examine the relation between EPU and the inbound cross-border deal flow.

Hypothesis 2: A higher EPU index is associated with decreased cross-border M&A inflow.

Moreover, institutional quality has been shown to affect significantly the M&A activity. Previous research has shown that the quality of institutions can affect various aspects of the M&A process, e.g. the deal completion time. Thus, it is only natural to assume that the quality of a country's institutions has an effect on the cross-border M&A activity, as acquirers are more likely to conduct deals where there is a higher level of institutional quality, because it can facilitate the whole deal process. Subsequently, the following hypothesis will explore the notion that EPU has a smaller effect in the cross-border M&A activity when a country presents higher institutional quality levels.

Hypothesis 3: The effect of EPU on the cross-border M&A activity is smaller in magnitude for countries with a higher degree of institutional quality than the respective effect for countries with a lower degree of institutional quality.

Lastly, M&A deals can be financed with different mixtures of cash, equity and even other financial assets, e.g. bonds, as mentioned previously. The choice between these different types of payment methods can be affected by country-specific factors, such as EPU. As cross-border deals are governed by higher information asymmetry levels than domestic deals, and EPU can lead to volatility in the future cash and financial constraints of a company, theoretically it should be less likely that bidders use liquid assets, i.e. cash, to finance deals when they face high EPU levels. Therefore, the following hypotheses, similar to the hypotheses tested by Nguyen & Phan (2017), will explore the relationship between EPU and the bidders' preferences for stock payments in cross-border M&A transactions, by utilizing the acquirer and target countries' EPU as the main explanatory variables.

Hypothesis 4: A higher EPU index in the acquirer country is positively associated with the bidders' preferences for stock payments in cross-border M&A deals.

Hypothesis 5: A higher EPU index in the target country is positively associated with the bidders' preferences for stock payments in cross-border M&A deals.

3. Methodology & data

The following section describes and presents the measurement methods of the variables, the data collection process, the models that will be used to investigate the different hypotheses developed before and the summary statistics of the variables.

3.1. Methodology

3.1.1. Economic policy uncertainty measurement

Prior literature on economic policy uncertainty has tried to approximate EPU in various different ways. For instance, the implied market volatility index, published by the Chicago Board of Options Exchange, which measures the standard deviation of stock returns and stock prices, is a widely accepted proxy for the uncertainty of the capital markets (Baker et al., 2016). However, this index is limited to identifying only the market uncertainty, and as per Al-Thaqeb & Algharabali (2019), it has little value in measuring uncertainty outside the markets.

Additionally, the FEARS index (Da et al., 2014) is composed by investigating the fear sentiments of investors from internet searches. Moreover, Scotti (2016) derives a “surprise factor” as a proxy for EPU, by measuring optimistic and pessimistic investor sentiments. Econometric techniques have also been proposed to create proxies for EPU. Jurado et al. (2015) use such techniques to create a proxy for macro-uncertainty focused on financial markets. The shortcoming of these methods is that they rely on the investors’ sentiments around macroeconomic events and do not specifically capture the policy aspect of EPU.

Baker, Bloom & Davis (2016) created the BBD index, which is considered a robust proxy of EPU, and it is preferred over other less comprehensive measures, such as national elections, the drawback of which is that they do not capture EPU in the periods between the elections, as national elections are usually held every four or five years. Numerous academics have already used the BBD index as a proxy for EPU (Bonaime et al., 2018; Sha et al., 2020; Nguyen & Phan, 2017; Aye et al., 2018; Li et al., 2021). This thesis will use the BBD index measured by the frequency of key terms in newspapers. This method scans newspapers for terms such as, or similar to, “economic” or “economy”, “uncertain” or “uncertainty” and “policy” or other policy-related terms. The advantage of this method is that it can be used in every country, adjusted for the respective language. Several other researchers have complemented the work of Baker, Bloom & Davis and created similar proxies for EPU, using the key-term frequency in newspapers method, for other countries. The BBD indices are publicly available in the economic policy uncertainty website developed by Baker, Bloom & Davis in 2012, with a monthly frequency. More information about the countries for which the BBD index is readily available, which act as the country sample of this thesis, is included in the appendices.

3.1.2. Model specifications

The hypotheses developed previously can be segmented into two distinct sets. The first set includes hypotheses one, two and three, exploring the relationship between EPU and cross-border M&A activity. The second set includes hypotheses four and five, investigating the bidders’ preferences for stock payments in cross-border deals in relation to EPU. Regarding the first set of hypotheses, previous researchers have used panel country-year observations to evaluate the relationship between EPU and cross-border M&A activity (Cao et al., 2017). In order to increase the observations in the country sample and to capture the short-term effect of EPU on the cross-border M&A activity, this thesis will use a quarterly frequency for the country level analysis. On the other

hand, cross-sectional models are regularly used to test the second set of hypotheses (Nguyen & Phan, 2017).

A number of scholars have used completed M&A deals (Sha et al., 2020), while numerous other researchers have used M&A announcements (Cao et al., 2017; Nguyen & Phan, 2017), as the dependent variables. As this thesis tries to examine the cross-border M&A activity, M&A announcements will be used as the dependent variable. The rationale is that companies have already engaged in M&A activity when they announce deals, even prior to that. EPU potentially affects the whole M&A process and especially the screening and target selection processes, thus M&A announcements provide a more comprehensive measure of the cross-border M&A activity.

3.1.2.1. Main explanatory variables

As described previously, the EPU, approximated by the BBD index, of the acquirer and target countries will act as the main explanatory variables. Usually, the BBD index is transformed into the natural logarithm of the three month (Nguyen & Phan, 2017), six month (Sha et al., 2020) or twelve month (Bonaime et al, 2018; Borthwick et al., 2020) average of the previous fiscal year. The findings of Nguyen & Phan (2017) are very similar for these three alternatives. Following Sha et al. (2020) this thesis will firstly use the six month transformation, because the six month period provides enough time for the markets to incorporate the effect of EPU, while also retaining a certain level of timeliness. Robustness tests will employ the three and twelve month transformations at a later stage. With regards to the country level analysis, which will include panel data with a quarterly frequency, the BBD index will be transformed into the natural logarithm of the average of the six months leading up to the quarter, lagged by one quarter, e.g. for the M&A announcements of the third quarter of 2015, the natural logarithm of the EPU average of the first six months (or two quarters) of 2015 will be utilized. For each deal announcement, regarding the deal level analysis, the BBD index will be transformed into the natural logarithm of the average of the last six months of the previous fiscal year, while robustness tests will employ the same transformation for the last three and twelve months of the previous fiscal year.

3.1.2.2. Control variables

Three sets of control variables will be employed, according to prior literature. The first set consists of country-specific control variables, regarding the target country, and is denoted by the vector C_{t-1} . As per existing literature, this set includes the annual growth rate of the gross domestic product

(GDP) and the natural logarithm of the gross domestic product per capita as proxies for the change in macroeconomic conditions and the level of economic development of each country (Cao et al., 2017), respectively. Another important factor in cross-border deals is the trade openness of each country. This factor is approximated by the percentage of the value of total imports and exports of a country to the country's GDP (Cao et al., 2017). Additionally, the stock market development and the access to financial institutions also play a significant role in cross-border investment decisions. In order to account for these factors, a variable representing the ratio of the stock market capitalization to the GDP of each country is included in the set. Moreover, country tax regulations can also affect the decision to engage in cross-border M&A activity. To approximate the tax regulations of a country, this thesis will employ the proportion of the country's tax revenue to the country's GDP, following Wang et al. (2014) and Li et al. (2021). Lastly, literature provides evidence that the quality of a country's institutions can affect significantly the M&A process. Following Lehne et al. (2014), the quality of institutions will be approximated by the sum of four out of the six World Bank Worldwide Governance Indicators (WGIs), i.e. the government effectiveness indicator, the regulatory quality indicator, the rule of law indicator and the control of corruption indicator, as the remaining two indicators, the voice and accountability indicator and the political stability and absence of violence indicator, reflect primarily the strength of political institutions. The aforementioned country-specific control variables are in line with previous literature (Cao et al., 2017), are available with a yearly frequency and will be lagged by one year, to eliminate unwanted biases in the data and auto-correlational effects, which could weaken the results (Cao et al., 2017).

The second set defines firm-specific control variables, denoted by the vector F_{t-1} , according to prior literature (Nguyen & Phan, 2017; Bonaime et al., 2018; Sha et al., 2020, Cao et al., 2017), which includes the control variables market-to-book ratio, firm size, ROA and leverage ratio of the bidder firm. These control variables will also be lagged by one year, as to account for the investor's (in this case the management team's) perspective and to eliminate other unwanted biases and auto-correlation effects which could weaken the results. The specific description of these variables is included in the appendices section.

The third set of control variables includes deal-specific control variables, denoted by the vector D . Firstly, a dummy variable is included to control for the status of the target, equal to 1 if the target is listed and to 0 otherwise, as the managerial motive theory proposes that the integration process

of a public company requires more time and effort (Draper & Paudyal, 2006). Secondly, the natural logarithm of the deal value will be employed, because it can significantly affect the choice of the type of payment, as deals with higher deal values are harder to be financed fully by cash. Lastly, a control variable indicating whether the two entities are in the same industry, based on the two-digit U.S. SIC codes, will be included in the sample, indicating whether the deal is diversifying or not. These control variables are in line with previous literature (Cao et al., 2017; Nguyen & Phan, 2017; Bonaime et al., 2018).

Finally, quarter and country fixed effects will be included in the country level analysis to remove the effects of macroeconomic conditions, such as oil price uncertainty or financial crises (Cao et al., 2017), and in order to account for potential omitted variable biases regarding various different effects in each country, respectively. Year, country (based on the target country) and industry (based on the two-digit U.S. SIC codes of the target firms) fixed effects will be included in the models exploring the relationship between EPU and the method of payment in cross-border deals.

3.1.2.3. EPU & cross-border M&A activity model

Regarding the first set of hypotheses, previous scholars have used the number of announced deals as the main explanatory variable (Cao et al., 2017), as mentioned before. This measure is log-normally transformed, in order to account for the non-normality between the activities in the various countries. More information about this transformation is provided in a later section of this thesis. The below model will employ panel country-quarter observations, i.e. with a quarterly frequency, according to previous literature (Cao et al., 2017), in order to examine the effect of EPU on the cross-border M&A activity:

$$(1) \quad LN(1 + \#of \text{ announced cross - border deals})_{i,t} = \beta_0 + \beta_1 * EPU_{i,t-1} + \beta_2 * C_{i,t-1} \\ + \gamma * Quarter_FE + \delta * Country_FE + \varepsilon_{i,t}$$

The model will have two specifications. The first one, applicable to the first hypothesis, will use the acquirer country EPU (and the announced outbound deals) as the main explanatory variable (dependent variable), while the second, attributable to hypothesis number two, will employ the target country EPU (and the announced inbound deals) as the main explanatory variable (dependent variable). Furthermore, in order to explore hypothesis number three, an interaction term, equal to the product of EPU and the quality of institutions control variable, will be added to the model. The model will use the pooled OLS regression method, and will include the country-specific control

variable vector C_{t-1} , lagged by one year, as described previously. Quarter and country fixed effects will also be included in this model. The independent variable EPU_{t-1} will be approximated by the natural logarithm of the six month average, i.e. the six months leading up to the previous quarter, of the BBD index. According to the findings of previous researchers, it is expected that high EPU levels decrease the inbound M&A flow (based on the delaying hypothesis) and increase the outbound M&A flow (based on the hedging hypothesis), as discussed in the hypotheses development section.

3.1.2.4. EPU & bidders' preferences for stock payments in cross-border M&As models

As mentioned previously, cross-border M&As tend to contain a greater level of information asymmetry, compared to domestic deals. High EPU could lead to higher volatility in the future cash flows and financial constraints of firms. This leads buyers to prefer to pay for the deals with their own stock, rather than exchanging liquid cash with the illiquid assets of the target. Hence, high EPU levels should lead to an increased likelihood of full stock payments in cross-border deals. Following a similar process to that utilized by Nguyen & Phan (2017), the below cross-sectional model will be employed to examine the relationship between EPU and the probability of full stock payments in cross-border M&A deals.

$$(2) \quad Stock_dummy_i = \beta_0 + \beta_1 * EPU_{t-1} + \beta_2 * C_{t-1} + \beta_3 * F_{t-1} + \beta_3 * D \\ + \gamma * Year_FE + \delta * Country_FE + \theta * Industry_FE + \varepsilon_i$$

The binary dummy variable *Stock_dummy* equals one if the payment is made fully in stock, and zero otherwise. A binary regression model should be used in order to investigate the likelihood, i.e. the probability, of a full stock payment. Various scholars have used a logit regression model (Bonaime et al., 2018; Borthwick, 2020; Li et al., 2021), while others preferred a probit regression model (Nguyen & Phan, 2017; Sha et al., 2020). These two models have insignificant differences between one another (Liao, 1994), so the choice between them will not affect significantly the results of the regressions. This thesis will use the logit regression model, as most of the previous literature has extensively used this model as well.

The model will have two specifications. The first one, corresponding to hypothesis number four, will employ the acquirer country EPU as the main explanatory variable, while the second, corresponding to the fifth hypothesis, will employ the target country EPU. The explanatory independent variable EPU_{t-1} will be approximated by the natural logarithm of the average of the

BBD index of the last six months of the previous fiscal year, as per existing literature. The model also includes the specified sets of control variables C_{t-1} , F_{t-1} & D , as described previously, and year, country and industry fixed effects (based on the target firm's country and industry).

Following Nguyen & Phan (2017), an alternative measure will be used as the dependent variable, in order to capture the effect of EPU on the percentage of stock included in the payment mix, as the original binary variable captures only the probability of full stock payments. Therefore, the dependent variable *Stock_proportion* will represent the proportion of stock used in the payment mix of cross-border M&As. As the proportion of stock in the payment mix can range from 0% to 100%, and the dependent variable is therefore bounded, a tobit regression model will be used, following previous literature (Nguyen & Phan, 2017). The aforementioned control variable vectors and year, country and industry fixed effects are also included in this model.

Robustness tests will include different specifications of the main explanatory variable, in order to examine the short and long-term effects of EPU. Additionally, as most of the cross-border M&A activity regards the U.S. and the U.K., robustness tests will remove these two countries from the country sample, and the deals which have these countries as the home or host countries from the deal sample, in an attempt to examine at what level the results are driven by the conditions in the aforementioned countries.

3.2. Data

The purpose of the following section is to describe the data collection and manipulation processes, as well as present sample summary statistics.

3.2.1. Data collection, data samples & databases

As mentioned before, the country sample includes the 24 countries for which the BBD index is available. The BBD index data are extracted from the Baker, Bloom & Davis website. The country-specific control variables are extracted from the World Bank online database.

The M&A deal sample consists of announced deals during the time period between 2010 and 2019, in order to exclude the effects of the 2008 financial crisis and the effects of the 2020 covid-19 pandemic. The M&A data are extracted from the Zephyr database, alongside with the corresponding deal and firm-specific control variables.

Following Erel et al. (2012), Cao et al. (2017) and Nguyen & Phan (2017) the following filters are applied to the M&A sample, leading to a valid cross-border M&A sample of 13,492 deals:

- The deal announcement date is between January 1, 2010 and December 31, 2019;
- The bidder and target countries belong to one of the 24 countries for which the BBD index is available (these countries act as the country sample of this thesis);
- Only cross-border merger and acquisition deals, as defined in the Zephyr database, are included in the sample, thus transactions such as LBOs, tender offers, recapitalizations, spin-offs, exchange offers, share repurchases and buybacks are not included in the sample;
- Cross-border M&A transactions with deal values below \$1 million are filtered out of the sample, as these deals are considered extraordinary small and might be influenced by various other factors which are not part of this paper's scope.

3.2.2. Data transformation

In order to complement the filters utilized in the data collection process and according to existing literature, the dataset described above is manipulated in order to optimize the analyses and remove unwanted biases.

Firstly, firm size is represented by the natural logarithm of the firm's total assets (Bonaime et al., 2018; Nguyen & Phan, 2017; Borthwick et al., 2020). The GDP per capita variable represents the natural logarithm of each country's respective value, in order to normalize the variable's distribution. The deal value variable, included in the deal level analysis, also represents the natural logarithm of the respective deal value, as to normalize the variable's distribution. Moreover, the data on firm size, market-to-book ratio, leverage ratio and ROA are winsorized in their first and ninety-ninth percentile values (Bonaime et al., 2018; Borthwick, 2020). Additionally, the country-specific data are also winsorised in the first and ninety-ninth percentile, in order to mitigate the effect of outliers, following Cao et al. (2017).

As described previously, the independent variable based on the BBD index is log-normally transformed (Nguyen & Phan, 2017; Bonaime et al., 2018; Sha et al., 2020). The dependent variables regarding the number of outbound and inbound announced cross-border deals per country are transformed into the natural logarithm of one plus the number of the respective deals, following Cao et al. (2017). An information table containing the specific description and the sources of all the variables included in this study is included in the appendices section.

3.2.3. Descriptive statistics

Table 3.2.3.1. presents the summary statistics of the variables for the respective samples described in the previous section. As mentioned earlier, firm and country-specific characteristics are winsorized on the 1st and 99th percentile, following previous literature.

Table 3.2.3.1.: Summary statistics table. The dependent variables outbound announcements (#) & inbound announcements (#) represent the natural logarithm of 1 + plus the number of announced deals. The EPU variables represent the natural logarithm of the 3, 6 and 12 month average, lagged by one quarter (regarding the country level analysis) or by one fiscal year (regarding the deal level analysis). All country & firm control variables are lagged by one year (or by one quarter where applicable).

<i>Variable</i>	<i>N</i>	<i>Mean</i>	<i>Min</i>	<i>Median</i>	<i>Max</i>	<i>Std. Dev.</i>
<i>Country sample</i>						
Outbound announcements (#)	960	2.102	0	2.1972	4.4886	1.1952
Inbound announcements (#)	960	2.2569	0	2.3026	4.5951	0.9655
EPU_3months	960	4.8826	3.0496	4.8391	6.6848	0.4683
EPU_6months	960	4.8839	3.1042	4.8455	6.5369	0.4474
EPU_12months	960	4.8842	3.1981	4.8541	6.3596	0.4238
Tax/GDP	880	0.1799	0.0905	0.1575	0.3382	0.0665
GDP growth	960	2.1139	-7.0867	2.079	10.6359	3.2102
GDP per capita	960	10.1726	7.2751	10.6183	11.1296	0.9242
Quality of institutions	960	4.2626	-2.6194	5.7771	8.3764	3.4511
Trade openness	960	0.9642	0.2393	0.6155	4.2598	0.9375
Market cap/GDP	960	1.2037	0.1854	0.7234	11.2471	2.0239
<i>Cross-border M&A deal sample</i>						
Stock_dummy	13,492	0.0423	0	0	1	0.2013
Stock_proportion	6,779	0.127	0	0	1	0.306
EPU_3months	13,492	5.0251	3.1488	4.9838	6.4131	0.4772
EPU_6months	13,492	5.0169	3.0714	4.9775	6.4533	0.4469
EPU_12months	13,492	5.0044	3.2959	4.9586	6.2967	0.4054
Firm size	9,181	6.2324	0.1401	6.4193	12.4982	2.7829
ROA	4,363	-0.043	-1.1516	0.0379	0.2832	0.2838
Leverage	4,364	0.2453	0	0.2145	0.9365	0.2073
Market/book	3,112	3.3458	0.3483	2.2923	23.8827	3.688
Deal value	13,492	3.7068	0.0862	3.5885	8.7608	1.978
Public target	13,492	0.1207	0	0	1	0.3258
Diversifying	13,492	0.468	0	0	1	0.499

The country sample consists of 960 country-quarter observations of 24 countries over 10 years (40 quarters), from 2010 to 2019. The EPU variables have similar mean values amounting to c.4.88 and to c.5, with regards to the country and deal samples respectively, with standard deviations ranging from c.0.4 to c.0.477. The variables *Stock_dummy* and *Stock_proportion* have mean values

amounting to c.0.0423 and c.0.127, respectively, meaning that on average 4.2% of the deals are fully financed with stock and that on average 12.7% of the deal value in cross-border M&A deals is financed with the bidder's stock.

Table 3.2.3.2.: Announced M&A deals per country. *The deals were announced during the 2010-2019 period.*

Country	Deals by acquirer country		Deals by target country	
	Number	Value (USD m)	Number	Value (USD m)
Australia (AU)	890	115,430	896	214,893
Belgium (BE)	208	310,147	253	64,881
Brazil (BR)	103	40,591	414	113,487
Canada (CA)	1,503	455,795	835	278,521
Chile (CL)	81	20,831	138	55,306
China (CN)	1,861	322,827	3,331	242,815
Colombia (CO)	53	15,084	114	28,438
Denmark (DK)	195	43,283	214	51,536
France (FR)	740	482,806	622	307,372
Germany (DE)	663	497,539	929	264,518
Greece (GR)	34	4,145	85	23,157
Hong Kong (HK)	2,040	274,132	1,805	249,192
India (IN)	274	42,912	562	105,189
Ireland (IE)	311	401,467	251	408,865
Italy (IT)	348	120,409	538	209,396
Japan (JP)	1,056	516,070	221	53,296
Mexico (MX)	96	40,216	178	48,461
Netherlands (NL)	763	607,424	622	263,070
Russia (RU)	159	102,326	388	107,476
Sweden (SE)	691	84,796	377	104,956
Singapore (SG)	1,401	213,867	811	125,335
Spain (ES)	332	137,582	540	255,077
U.K.(GB)	2,194	804,202	1,912	1,264,692
U.S.A.(US)	3,072	1,525,623	3,423	2,259,492

Table 3.2.3.2. portrays the announced deals per country during the sample period. The U.K. and the U.S. account for c.27.5% of the deal sample, regarding both the deals per acquirer and target countries, while deals targeting China account for c.17% of the sample. The lowest deals per acquirer country are presented in Greece and Colombia, with 34 and 53 deals respectively, while the lowest deals per target country are presented again in Greece and Colombia, with 85 and 114 deals respectively.

The Pearson correlation coefficients are included in the appendices section, as the results of these tests should not be used to extract conclusions about the relationship between variables, because they do not account for the effect of the control variables nor the different types of relationships between them. Even though no strong correlations between the variables are visible from the Pearson correlation results, potential multicollinearity between the variables will be tested in the next stage of this study, using the variable inflation factors. It should be noted that the three main explanatory variables, i.e. the three, six and twelve month EPU transformations, are almost perfectly correlated with each other, suggesting that the regression results in the robustness tests should be relatively similar to each another, as also noted by Nguyen & Phan (2017).

4. Results

This section discusses the regression results in the context of the hypotheses developed earlier and in comparison to the results of previous literature.

The variance inflation factors (VIF) were used to test for potential multicollinearity in the models. As all the VIFs presented values lower than five, no strong correlation exists between the explanatory variables, as also anticipated from the results of the Pearson correlation tests.

4.1. Country level analysis

Table 4.1.1. illustrates the results of the pooled OLS regression of the country-quarter panel model, exploring the relation between EPU and the cross-border M&A activity. Panel A depicts the estimation results on the outbound cross-border M&A activity, while Panel B presents the estimation results on the inbound cross-border M&A activity. The dependent variable is the natural logarithm of one plus the number of announced outbound and inbound deals per quarter, regarding Panel A and Panel B respectively. All control variables are lagged by one quarter (or by one year regarding the country-specific control variables, as they are available only with a yearly frequency). The model includes country and quarter fixed effects (the coefficients of which have been suppressed). Following Cao et al. (2017) the standard errors are adjusted for heteroscedasticity and have been clustered at the country level, as the t-statistics (which are reported in the parentheses) need to be adjusted for within-country correlation. Different specifications excluding the control variable Tax/GDP are presented, as the missing values in the abovementioned variable drop the valid number of observations to 880. In order to explore hypothesis number three, an interaction term equal to the product of the main explanatory variable and the quality of institutions control variable has been included in specifications (3) and (4) in both panels.

Columns (1) and (2) in Panel A show a negative relationship between EPU and the outbound cross-border M&A activity. Both columns depict negative, but not statistically significant, estimated coefficients for the EPU index equal to -0.0998 and -0.066 respectively, economically meaning that a one standard deviation increase in EPU (0.4474) is associated with estimated decreases of 4.46% (0.0998×0.4474) and 2.95% (0.066×0.4474), respectively for columns (1) and (2), in the outbound cross-border M&A activity. Overall, the aforementioned results cannot support hypothesis number one, under this specific methodology and sample, and contradict the respective results of Cao et al. (2017), who found a positive and statistically significant relationship between

Table 4.1.1.: Country-quarter pooled OLS regression. Panel A depicts the regression analysis regarding hypothesis number 1, while Panel B depicts the regression analysis concerning hypothesis number 2. The dependent variables are presented at the top, equal to $\ln(1 + \text{number of announced cross border deals})$. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	Panel A				Panel B			
	<i>LN(1+Outbound # of cross-border M&A deals)</i>				<i>LN(1+Inbound # of cross-border M&A deals)</i>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
EPU_6months	-0.0998 (-0.9918)	-0.0660 (-0.6672)	-0.1122 (-0.6597)	-0.0751 (-0.4707)	-0.1751*** (-2.8321)	-0.1857*** (-3.1367)	-0.2297** (-2.3730)	-0.2149** (-2.2817)
Tax/GDP		0.4822 (0.2380)		0.4759 (0.2385)		1.4757 (0.5968)		1.4555 (0.5982)
Market cap/GDP	0.1843 (0.9478)	0.5178* (1.9828)	0.1872 (0.9691)	0.5187* (1.9917)	0.0023 (0.0275)	0.0874 (0.6557)	0.0149 (0.1818)	0.0903 (0.6865)
Quality of institutions	0.0360 (0.4427)	-0.0032 (-0.0385)	0.0158 (0.0823)	-0.0184 (-0.1064)	-0.0666 (-0.7307)	-0.0846 (-0.8449)	-0.1553 (-1.2562)	-0.1332 (-1.0415)
EPU * Quality of institutions			0.0040 (0.1532)	0.0031 (0.1242)			0.0176 (1.0607)	0.0098 (0.6132)
GDP growth	0.0161* (2.0650)	0.0140 (1.7016)	0.0157** (2.2039)	0.0136 (1.7021)	0.0191** (2.3744)	0.0195** (2.4224)	0.0173** (2.1114)	0.0183** (2.2149)
GDP per capita	0.1635 (0.4990)	0.2946 (0.8489)	0.1645 (0.4998)	0.2954 (0.8479)	0.0353 (0.1716)	-0.0476 (-0.2371)	0.0398 (0.1952)	-0.0451 (-0.2242)
Trade/GDP	-0.2132** (-2.1118)	-0.3313* (-1.9599)	-0.2078** (-2.2118)	-0.3228* (-1.9174)	-0.5422** (-2.1525)	-0.6109 (-1.4380)	-0.5185** (-2.1342)	-0.5840 (-1.3578)
Constant	0.7222 (0.1975)	-0.8917 (-0.2313)	0.7665 (0.2015)	-0.8610 (-0.2179)	3.5167 (1.6455)	4.1430* (1.9389)	3.7115* (1.8530)	4.2413* (2.0749)
N	960	880	960	880	960	880	960	880
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0254	0.0417	0.0245	0.0407	0.0284	0.0353	0.0295	0.0349

EPU and the outbound cross-border activity. These results suggest that the hedging hypothesis does not govern the outbound cross-border M&A decision of companies. Outbound cross-border M&As seem to be delayed, even though this effect is not statistically significant, when companies are facing high EPU levels in their home countries, meaning that the delaying hypothesis may affect outbound M&A activity, as firms seem to prefer a certain level of stability in their home countries when considering the irreversible costs of engaging in outbound cross-border M&As.

Columns (1) and (2) in Panel B show a negative relationship between EPU and the inbound cross-border M&A activity. Specifically, both columns depict negative and statistically significant, at the 0.01 level, estimated coefficients for the EPU index, amounting to -0.1751 and -0.1857 respectively. Economically, a one standard deviation increase in the EPU index (0.4474) is associated with estimated decreases of 7.8% (0.1751×0.4474) and 8.3% (0.1857×0.4474), regarding specifications (1) and (2) respectively, in the inbound cross-border M&A activity. These

results are both statistically and economically significant and hence hypothesis number two can be empirically supported. The abovementioned results are similar to those of Cao et al. (2017) regarding the relationship between EPU and the inbound M&A activity. Hence, there are valid grounds to support the delaying hypothesis as a potential explanation for the negative relationship between EPU and the inbound cross-border M&A activity.

As described previously, an interaction term, denoted as “EPU * Quality of institutions”, calculated as the product of the EPU index and the quality of institutions variable, is included in specifications (3) and (4) in both panels in order to test hypothesis number three, i.e. that the effect of EPU in cross-border M&A activity is smaller in countries with a higher level of institutional quality. Specifications (3) and (4) in Panel A illustrate negative and not statistically significant estimated coefficients for EPU amounting to -0.1122 and -0.0751 respectively. Accordingly, the positive and not statistically significant estimated coefficients for the interaction term amount to 0.004 and 0.0031, regarding columns (3) and (4) respectively. Economically, this means that when the quality of institutions variable is held constant, a one standard deviation increase in EPU (0.4474) would decrease the outbound M&A activity by 4.8% ($0.4474 \times (-0.1122 + 0.004)$) and by 3.2% ($0.4474 \times (-0.0751 + 0.0031)$), concerning specifications (3) and (4) respectively. The fact that the interaction terms are positive mean that the negative effect of EPU on the outbound activity becomes smaller as the quality of institutions variable increases, supporting the theoretical framework of hypothesis number three. Despite that, as the results are not statistically significant, hypothesis number three cannot be empirically accepted, under this proposed methodology.

Continuing with Panel B, specifications (3) and (4) depict negative and statistically significant, at the 0.05 level, estimated coefficients for the main explanatory variable, amounting to -0.2297 and -0.2149 respectively. The interaction terms are positive, but not statistically significant, and amount to 0.0176 and 0.0098 regarding columns (3) and (4) respectively. Economically, this means that when the quality of institutions variable is held constant, a one standard deviation increase in EPU (0.4474) would decrease the inbound M&A activity by 9.5% ($0.4474 \times (-0.2297 + 0.0176)$) and by 9.2% ($0.4474 \times (-0.2149 + 0.0098)$), concerning specifications (3) and (4) respectively. The positive signs of the interaction terms mean that the negative relationship between the EPU and the inbound M&A activity becomes less prominent when the country’s institutional quality increases, supporting again the theoretical framework behind hypothesis number three. However, as no

statistical significance could be established for the interaction terms under this proposed model, hypothesis number three is once again empirically rejected.

These results regarding the models including the interaction term are not surprising, as the standalone quality of institutions variable is not statistically significant and in most cases it presents a negative relationship with cross-border M&A activity. The most probable explanation is that there are issues with the measurement method of the quality of institutions control variable, as various previous researchers have used alternative measures from different data sources. An analysis including the standalone constituents of the quality of institution variable is included in the appendices section. The results regarding the interaction terms are insignificant for the standalone variables, which comprise the quality of institutions variable, indicating that there are probably severe measurement issues with the quality of institutions variable.

4.2. Cross-border M&A deal analysis

Table 4.2.1. presents the results of the logit and tobit regressions regarding hypotheses four and five, exploring the relationship between EPU and the method of payment in cross-border M&A deals. It should be noted that the columns regarding the logit regression models present the logit marginal effects and not the logit coefficients, in order to facilitate the interpretation of the results (the original logit coefficients are presented in the appendices). The dependent variables *Stock_dummy* and *Stock_proportion* correspond to the logit and tobit models respectively. The models include country, industry (based on the target firm's country and industry) and year fixed effects (the coefficients of which have been suppressed), while the standard errors have been adjusted for heteroscedasticity and are clustered by the acquirer country, following Cao et al. (2017), in order to account for autocorrelation effects in the acquirer's country.

Panel A underlines the analysis regarding the relation between the method of payment and the acquirer country EPU. Column (1) depicts a positive, but not statistically significant, estimated marginal effect amounting to 0.0148 for the main explanatory variable, meaning that a one standard deviation increase in the EPU variable (0.4469) would result in an estimated increase of 1.48% in the probability of a full stock payment in the average cross-border deal, which cannot be considered economically meaningful. Specification (2) depicts a positive, but not statistically significant, estimated tobit coefficient of 0.0418. Economically this translates in a slightly meaningful

Table 4.2.1.: Cross sectional logit & tobit regression analyses exploring the relationship between EPU & the method of payment in cross-border M&A deals. The dependent variable in specification (1) is the *Stock_dummy*, while in specification (2) the *Stock_proportion*. The z-statistics (t-statistics) presented in the parentheses for the logit (tobit) model have been calculated based on heteroscedasticity-robust standard errors clustered by acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

<i>Variables</i>	Panel A		Panel B	
	<i>Acquirer country EPU</i>		<i>Target country EPU</i>	
	<i>Logit marginal effects</i> (1)	<i>Tobit</i> (2)	<i>Logit marginal effects</i> (1)	<i>Tobit</i> (2)
EPU_6months	0.0148 (0.9321)	0.0418 (1.6055)	0.0416*** (2.7050)	0.0410*** (3.9195)
<i>Firm characteristics</i>				
Firm size	-0.0212*** (-11.0761)	-0.0532*** (-8.4513)	-0.0222*** (-14.4320)	-0.0570*** (-5.5046)
ROA	-0.0371*** (-6.7547)	-0.2942*** (-5.5819)	-0.0165 (-1.3873)	-0.1907*** (-2.6505)
Leverage	-0.0456 (-1.2980)	-0.0637 (-1.3315)	-0.0303 (-0.7380)	-0.0168 (-0.2472)
Market/Book	-0.0012 (-0.9719)	0.0007 (0.2500)	-0.0010 (-1.0259)	-0.0001 (-0.0454)
<i>Country characteristics</i>				
Tax/GDP	-0.0825 (-0.1409)	0.5406 (1.1120)	0.0247 (0.3795)	0.2483* (1.7000)
Trade/GDP	-0.1450* (-1.7504)	-0.2531 (-1.3806)	0.0118 (0.9797)	0.0183 (1.5326)
GDP growth	-0.0021 (-1.6005)	-0.0004 (-0.1006)	-0.0025 (-1.2442)	-0.0022 (-0.6725)
GDP per capita	-0.0354 (-0.6266)	-0.1271 (-1.3688)	0.0281** (2.1084)	0.0092 (0.4072)
Quality of institutions	-0.0055 (-0.2894)	0.0517 (1.6231)	-0.0117*** (-2.6953)	-0.0135 (-1.3813)
Market cap/GDP	-0.0120 (-0.2647)	-0.0413 (-0.5829)	0.0184 (0.9122)	0.0362 (1.2621)
<i>Deal characteristics</i>				
Deal value	0.0107*** (2.8720)	0.0329*** (8.9324)	0.0100*** (3.6498)	0.0304*** (8.4331)
Public target	0.0597*** (4.8490)	0.0729** (2.2973)	0.0581*** (6.6910)	0.0803*** (2.6636)
Diversifying	-0.0070 (-0.9254)	-0.0260 (-1.3139)	0.0181 (1.5956)	0.0197** (2.1886)
Constant	N/A N/A	1.4886 (1.1861)	N/A N/A	0.4300* (1.8445)
N	2,261	1,448	2,288	1,448
Country fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Pseudo R ²	0.372	0.602	0.380	0.610

estimated increase of 1.9% (0.0418×0.4469) in the percentage of stock used in the payment mix of cross-border deals, when the acquirer country EPU increases by one standard deviation (0.4469). Even though these estimated positive relationships support the underlying theoretical framework of hypothesis number four, i.e. that higher levels of EPU in the acquirer country should be positively related to the bidders' preferences for stock payments, hypothesis number four cannot be empirically accepted, as no statistical significance could be established under the specific methodologies and sample.

Panel B illustrates the analysis regarding the relation between the method of payment and the target country EPU. Column (1) presents a positive and statistically significant, at the 0.01 level, estimated marginal effect amounting to 0.0416 for the EPU variable. Economically this means that a one standard deviation increase in the EPU variable (0.4469) would result in an estimated increase of 4.16% in the probability of a full stock payment in the average cross-border deal, which can be considered economically significant. Column (2) depicts a positive and statistically significant, at the 0.01 level, estimated tobit coefficient of 0.0410 regarding the EPU variable. Economically this translates into an estimated increase of 1.83% (0.0410×0.4469), which is considered a somewhat economically meaningful effect, in the percentage of stock used in the payment mix of cross-border deals, when the target country EPU increases by one standard deviation (0.4469). These estimated positive relationships support the underlying argument of hypothesis number five, i.e. that higher levels of EPU in the target country should lead to bidders preferring stock-based payments over other methods of payment. Moreover, hypothesis number five can be empirically supported, as the aforementioned positive relationships are statistically significant, under these specific methodologies and sample.

The aforementioned results regarding the deal level analysis provide significant empirical indications that the asymmetric information problem governs the choice of the method of payment in cross-border deals when the target country's EPU is high, and that bidders are reluctant to exchange liquid cash for illiquid assets in high uncertainty periods. Furthermore, the positive relationships found in both panels are in line with the results of Nguyen & Phan (2017), who found a positive and statistically significant relationship between domestic EPU and the stock payment likelihood, and stock proportion, in their sample of intra-U.S. domestic M&A deals.

4.3. Robustness of results

This part of the thesis illustrates the robustness tests corresponding to the regression analyses presented in the previous section. Regarding the endogeneity concerns of this study, as the EPU explanatory variable is part of the macro-level data and the cross-border M&A deals (regarding the announced number of deals and the method of payment in these deals) are part of the of the micro-enterprises level data, Li et al. (2021) state that the reverse causality study of the economic policy uncertainty in cross-border deals can be ignored. Moreover, the fact that the control variables are lagged provides some level of comfort regarding any auto-correlation problems and avoids unwanted biases in the data sample. Lastly, the number of control variables can be considered large enough, especially in the deal level analysis, to provide a relative level of comfort regarding the omitted variable bias.

4.3.1. Country level analysis

Table 4.3.1.1. presents the first robustness test regarding the country level analysis, which includes the alternative measures of the main explanatory variable, i.e. the EPU index is transformed into the natural logarithm of the average of the three and twelve months leading up to a quarter (lagged

Table 4.3.1.1.: Country level pooled OLS robustness analysis including the alternative measures of the main explanatory variable. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level. Country-specific control variables are suppressed for simplicity reasons.

Variables	Panel A				Panel B			
	LN(1+Outbound # of cross-border M&A deals)				LN(1+Inbound # of cross-border M&A deals)			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
EPU_3months	-0.0445 (-0.5197)	-0.0369 (-0.2704)			-0.1336*** (-2.9014)	-0.1750** (-2.2956)		
EPU_12months			-0.0969 (-0.8623)	-0.1353 (-0.7237)			-0.2074** (-2.5582)	-0.2101 (-1.6896)
EPU * Quality of institutions		-0.0026 (-0.1216)		0.0130 (0.4482)		0.0139 (1.0424)		0.0009 (0.0431)
Country characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	880	880	880	880	880	880	880	880
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0409	0.0398	0.0432	0.0430	0.0289	0.0295	0.0359	0.0348

by one quarter), in an attempt to compare the short and long-term effects of EPU on the cross-border M&A activity, to the mid-term effect examined in the previous section. Panel A (Panel B) depicts the relationship between EPU and the outbound (inbound) cross-border M&A activity.

As can be seen in the above table, the results are similar to the original results. All specifications in Panel A depict negative, and not statistically significant, estimated coefficients for the EPU measures, similar to the estimated coefficients of the original tests. This means that hypothesis number one is once again both theoretically and empirically rejected, meaning that the delaying, and not the hedging, hypothesis most likely governs the relation between EPU and the outbound M&A activity. Specifications (1), (2) and (3) in Panel B show negative, and statistically significant, estimated coefficients regarding the EPU variable, hence supporting empirically the underlying argument behind hypothesis number two. However, in contrast with the original results, no statistical significance could be established under specification (4). Moreover, the effects of the EPU variables on the inbound activity become increasingly more prominent as the time period increases, i.e. the economic effect of the three month transformation presented in specification (1) is the smallest ($-0.1336 \times 0.4683 = -6.3\%$), followed by the original six month measure (-8.3%) and by the twelve month alternative depicted in column (3) ($-0.2074 \times 0.4238 = -8.8\%$), indicating that the longer the high EPU levels affect a country, the larger the negative effect on the cross-border inbound activity. The results for the specifications including the interaction term in both panels are again not statistically significant and cannot support hypothesis number three.

The next robustness test regarding the country level analysis removes from the country sample the U.S. and the U.K., as most of the deal activity historically originates from these two countries, in an attempt to examine at what level the results are driven by the conditions in these two countries. Table 4.3.1.2. depicts the estimated coefficients of the pooled OLS model, while Panel A (Panel B) illustrates the analysis attributable to the outbound (inbound) cross-border M&A activity, where the dependent variable is the natural logarithm of one plus the number of announced deals in each country, with a quarterly frequency. The results in both panels are in line with the original results, i.e. they depict negative and not statistically significant estimated coefficients for the EPU variable in Panel A, rejecting hypothesis number one. Furthermore, Panel B depicts negative and statistically significant estimated coefficients for all three alternative measures of EPU, supporting hypothesis number two. Additionally, the negative estimated effect of EPU on the inbound cross-border M&A activity becomes more prominent the longer the high EPU levels affect a country, meaning that the EPU estimated coefficient for the three month transformation is lower than the one for the six month transformation, and in turn the coefficient for the twelve month transformation is the largest one out of the three. Hypothesis number three is once again rejected

regarding both the outbound and the inbound analyses, as no statistical significance could be established in any specification for the interaction term. These results indicate that even though the U.S. and the U.K. historically drive the global M&A activity, the estimated relationships between the EPU and the cross-border M&A activity are robust when these two countries are not included in the country sample.

Table 4.3.1.2.: Country level pooled OLS robustness analysis excluding the U.S. & the U.K. from the country sample. Panel A (Panel B) illustrates the regression results attributable to the outbound (inbound) cross-border M&A activity. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level. Country-specific control variables are suppressed for simplicity reasons.

Variables	Panel A						Panel B					
	LN(1+Outbound # of cross-border M&A deals)						LN(1+Inbound # of cross-border M&A deals)					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
EPU_3months	-0.0021 (-0.0229)	-0.0088 (-0.0686)					-0.1505** (-2.8509)	-0.1822** (-2.3366)				
EPU_6months			-0.0201 (-0.1893)	-0.0401 (-0.2686)					-0.2093*** (-3.0316)	-0.2265** (-2.3274)		
EPU_12months					-0.0517 (-0.4223)	-0.0939 (-0.5344)					-0.2268** (-2.3574)	-0.2241* (-1.7569)
EPU * Quality of institutions		0.0027 (0.1329)		0.0083 (0.3399)		0.0181 (0.6204)		0.0129 (0.8861)		0.0071 (0.4022)		-0.0011 (-0.0467)
Country characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	800	800	800	800	800	800	800	800	800	800	800	800
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0591	0.0580	0.0593	0.0585	0.0600	0.0603	0.0266	0.0267	0.0334	0.0325	0.0326	0.0314

In conclusion, the robustness results regarding the country level analysis are similar to the original results, i.e. they do not empirically support hypotheses one and three, but they do provide empirical evidence supporting hypothesis number two, while the relationship under question becomes more prominent when a country is experiencing relatively higher levels of EPU for longer time periods.

4.3.2. Deal level analysis

Table 4.3.2.1. portrays the first robustness test regarding the deal level analysis, examining the relationship between EPU and the method of payment in cross-border M&A deals, which includes different specifications of the main explanatory variable. Specifically, the EPU index in this case is transformed into the natural logarithm of the average of the last three and twelve months of the previous fiscal year, in an attempt to compare the short and long-term effects of EPU on the method of payment to the mid-term effect examined in the previous section. As it can be seen from the two regression panels, the robustness results are very similar to the original results, as expected from the almost perfect correlation between the three explanatory EPU measures and from the results of Nguyen & Phan (2017), who found insignificant differences between the three measures, as

mentioned before. Panel A (Panel B) denotes the analysis exploring the relationship between the acquirer (target) country EPU and the method of payment, while specifications (1) and (2) present the estimated logit marginal effects of the logit model utilizing the full stock payment dummy as the dependent variable, and specifications (3) and (4) present the estimated tobit coefficients utilizing the proportion of stock included in the payment mix of the deals as the dependent variable. The original logit coefficients are included in the appendices section.

Table 4.3.2.1.: Cross sectional logit & tobit robustness regressions using different measures of EPU regarding hypothesis number four and five. The dependent variable for specifications (1) & (2) is the *Stock_dummy*, while for (3) & (4) is the *Stock_proportion*. The main explanatory variable in Panel A (Panel B) is the acquirer (target) country EPU. The z-statistics (t-statistics) presented in the parentheses for the logit (tobit) model have been calculated based on heteroscedasticity-robust standard errors, clustered by the acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level. The control variable coefficients are presented only in the appendices section.

Variables	Panel A Acquirer country EPU				Panel A Target country EPU			
	Logit marginal effects		Tobit		Logit marginal effects		Tobit	
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
EPU_3months	0.0089 (0.5913)		0.0349 (1.3448)		0.0312* (1.9566)		0.0302*** (3.0539)	
EPU_12months		0.0185 (1.0172)		0.0518 (1.6258)		0.0471*** (3.4690)		0.0496*** (6.4903)
Firm characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country characteristic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,261	2,261	1,448	1,448	2,288	2,288	1,448	1,448
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.372	0.372	0.692	0.692	0.376	0.383	0.696	0.703

The results in Panel A are very similar to the original results, illustrating positive, but not statistically significant, estimated coefficients for the alternative measures of EPU, supporting the theoretical framework of hypothesis number four, but as no statistical significance could be established, hypothesis number four cannot be empirically accepted under this robustness test.

The results in Panel B are also similar to the original results, illustrating positive and statistically significant estimated coefficients for the alternative measures of EPU, supporting both theoretically and empirically hypothesis number five. Economically, a one standard deviation increase in the three month EPU variable (0.4772) would lead to estimated increases of 3.12% and 1.44% (0.4772*0.0302) in the likelihood of a full stock payment for the average deal (specification (1))

and in the proportion of stock used in the payment mixture of cross-border deals (specification (3)), respectively. Furthermore, a one standard deviation increase in the twelve month EPU variable (0.4054) would lead to estimated increases of 4.7% and 2% (0.4054×0.0496) in the likelihood of a full stock payment for the average cross-border deal (specification (2)) and in the proportion of stock used in the payment mix of cross-border deals (specification (4)), respectively.

These results indicate that the long-term EPU has a greater effect on the choice of payment in cross-border deals, than the short-term EPU transformation, while the original mid-term EPU variable has an estimated economical effect which is between the short-term and the long-term effects. The underlying reasoning is that the longer the duration of the high EPU climate in the target country, the largest its effect on the choice between payment methods.

To sum up, the abovementioned robustness results are very similar to those of previous scholars and closely follow the original outcomes of the deal level analysis, meaning that higher levels of short and long-term EPU in both the acquirer and target country lead companies to prefer to use their own equity as a method of payment, as they are reluctant to exchange liquid cash for other illiquid assets of the target.

The last robustness test regarding the deal level analysis removes from the sample the deals that have the U.S. and the U.K. as the host or home countries. Most of the deal activity historically originates from these two countries, thus this test is an attempt to examine at what level the original results are driven by the conditions in these two countries. Table 4.3.2.2. depicts the regression results including all three transformations of the main explanatory variable. Panel A (Panel B) illustrates the analyses regarding hypothesis number four (hypothesis number five). Specifications (1), (2) and (3) depict the estimated logit marginal effects of the logit regressions where the dependent variable is the *Stock_dummy*, while specifications (4), (5) and (6) depict the estimated tobit coefficients of the regressions where the dependent variable is the *Stock_proportion*. The results in Panel A are similar to the original results. Specifically, specifications (1) through (3)

depict positive, but not statistically significant estimated coefficients for the main explanatory variables.

Table 4.3.2.2.: Cross sectional logit & tobit robustness regression results excluding the deals that have the U.S. & the U.K. as the host or home country. The dependent variable for specifications (1), (2) & (3) is the *Stock_dummy*, while for (4), (5) & (6) the *Stock_proportion*. The main explanatory variable in Panel A (Panel B) is the acquirer (target) country EPU. The z-statistics (t-statistics) presented in the parentheses for the logit (tobit) model have been calculated based on heteroscedasticity-robust standard errors, clustered by the acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level. The control variable coefficients are presented only in the appendices section.

Variables	Panel A Acquirer country EPU						Panel B Target country EPU					
	Logit marginal effects			Tobit			Logit marginal effects			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
EPU_3months	0.0335 (0.6764)			0.0709* (1.6899)			0.0365** (2.3061)			0.0481*** (2.9121)		
EPU_6months		0.0448 (0.9399)			0.0754* (1.7809)			0.0588*** (4.5100)			0.0577*** (3.1020)	
EPU_12months			0.0421 (0.8204)			0.0905* (1.9149)			0.0666*** (7.3741)			0.0702*** (4.5624)
Firm characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	619	619	619	530	530	530	674	674	674	530	530	530
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.499	0.500	0.499	0.839	0.840	0.839	0.491	0.499	0.502	0.831	0.839	0.842

However, the tobit models in this case lead to positive and statistically significant, at the 0.1 level, estimated coefficients, in columns (4) through (6), meaning that hypothesis number four is empirically supported under this methodology, according to the results of existing literature. It should be noted that this is the only instance that hypothesis number four is statistically supported in this study. Panel B depicts results similar to those of the original analysis, i.e. positive and statistically significant estimated coefficients for the EPU variables, regarding all specifications. This means that the original results regarding hypothesis number five are robust when the deals where the U.S. and the U.K. act as the host or home countries are removed from the sample.

Overall, the original results regarding hypotheses number four and five are robust when the short and long-term EPU explanatory variables are utilized in the models. Furthermore, the effect of EPU on the preference of the bidders for stock payments becomes more prominent when the acquirer and target countries are affected from high EPU levels for longer time periods. Lastly, the original results are robust when the deals where the U.S. and the U.K. act as the home or host countries, which are historically responsible for the majority of the cross-border M&A activity, are removed from the deal sample.

5. Shortcomings & limitations

It must be noted that this study is restricted by a number of shortcomings and limitations, with regards to methodological issues, variable measurement issues and data sample restrictions.

Firstly, this thesis follows closely the proposed methodology of Cao et al. (2017) to explore the relation between EPU and cross-border M&A activity. This methodology emphasizes on the cross-border deal flow, by using the number of announced cross-border transactions. However, that might not be the optimal methodology, as other researchers have proposed logit models emphasizing on the likelihood of engaging in M&A activity, by using dummy binary variables as the dependent variable (Nguyen & Phan, 2017). Moreover, the choice of announced deals is by itself an influential factor, as other researchers have preferred completed deals. Secondly, even though the BBD index is regarded as a comprehensive measure of EPU by numerous researchers, it has some limitations. For instance, the measurement method of the BBD index is subject to certain constraints relating to the available newspaper sources in each country. One important factor is the level of the freedom of speech in each country, as it can heavily influence the critical terms used in news articles relating to the economic policy uncertainty. Furthermore, other measures of EPU, such as national elections, could have a different impact on the outcomes of this research.

Thirdly, even though this study employs several control variables, regarding firm-specific, country-specific and deal-specific characteristics, there might be a number of additional factors affecting the relationship between cross-border M&A activity and EPU. For example, there are evidence that the cultural and religious distances can affect cross-border transactions between country pairs. Additional firm characteristic could also affect the decision to engage in M&A activities, such as the stock price progression of the bidder company prior to the M&A announcement.

Fourthly, M&A activity is known to be affected by various incentives of the management teams of the bidder firms. The compensation of said management teams is very often tied to the completion of M&A deals in the form of deal bonuses. In addition, the empire building theory suggests that managers may engage in M&As to increase their own power and serve their interest, at the expense of the shareholders, while not taking into account whether the deal is beneficial or not, leading to irrational M&A investment activity, which could potentially affect the data sample used in this study. Lastly, the results might be influenced by the selected sample period, hence different sample periods may yield different outcomes.

6. Conclusions

This thesis examines the relationship between economic policy uncertainty, approximated by the Baker, Bloom and Davis index, and the cross-border M&A activity by using a country sample of 24 countries over the 2010-2019 period, as well as the effect of EPU on the method of payment in cross-border M&A deals, by using a sample of announced cross-border M&As over the same period.

Based on the hedging hypothesis, Cao et al. (2017) provided evidence that the outbound cross-border M&A activity is positively related to EPU, as approximated by national elections. This thesis found a negative, but not statistically significant, relationship between EPU and the outbound cross-border M&A activity, suggesting that the delaying hypothesis governs the relationship between EPU and the outbound M&A activity, as companies seem to prefer a certain level of certainty in their home countries when considering their international investments. Additionally, based on the delaying hypothesis, Cao et al. (2017) provided evidence that the inbound cross-border M&A activity is negatively related to EPU. This study provided similar significant evidence supporting empirically the hypothesis that bidders delay their cross-border M&A investments when they are facing high EPU levels in the target country.

Moreover, this thesis tried to contribute to previous literature by examining how the quality of institutions can affect the relationship between EPU and cross-border M&A activity, by including an interaction term equal to the product of the EPU and the quality of institutions variable in the models. The results cannot empirically support the hypothesis that the effects of EPU on the cross-border M&A activity are smaller for countries with higher levels of institutional quality, as no statistically significant relationship could be established for the aforementioned interaction term.

Furthermore, this study also tried to examine how EPU affects the method of payment in cross-border M&A deals, by using a similar methodology to the one utilized by Nguyen & Phan (2017), who however only focused on domestic U.S. deals. Based on the general accepted notion that cross-border deals are governed by higher levels of asymmetric information, bidders should in theory be discouraged to exchange liquid cash with illiquid assets, such as the target's stock, when they face high EPU levels, and should instead prefer to fund these cross-border deals with their own stock. A significant positive relationship between EPU and the bidders' preferences for stock payments in cross-border M&A deals was established, supporting the results of previous literature.

7. References

- Agrawal, A., Jaffe, J. F., & Mandelker, G. N., (1992), The post-merger performance of acquiring firms: a re-examination of an anomaly, *The Journal of Finance*, 47(4), 1605-1621.
- Ahern, K. R., Daminelli, D., & Fracassi, C. (2015), Lost in translation? The effect of cultural values on mergers around the world, *Journal of Financial Economics*, 117, 165–189.
- Alexandridis, G., A. Antoniou, and H. Zhao (2008), Belief Asymmetry and Gains from Acquisitions, *Journal of Multinational Financial Management* 18: 443–460.
- Al-Thaqeb, S. A., & Algharabali, B. G. (2019), Economic policy uncertainty: A literature review, *The Journal of Economic Asymmetries*, 20(e00133), 1–11.
- Andrade, G., Mitchell, M. & Stafford, E., (2001), New evidence and perspectives on mergers, *Journal of Economic Perspectives*, 15, pp 103–120.
- Aye, G. C., Balcilar, M., Demirer, R., & Gupta, R. (2018), Firm-level political risk and asymmetric volatility, *The Journal of Economic Asymmetries*, 18(e00110), 1–7.
- Baker, S. R., N. Bloom, and S. J. Davis, (2016), Measuring Economic Policy Uncertainty, *Quarterly Journal of Economics*, 131, no. 4 (November), 1593-1636.
- Baker, S.R., N. Bloom, S.J. Davis and Xiaoxi Wang, (2013), Economic Policy Uncertainty in China, unpublished paper, University of Chicago.
- Barber, B.M. & Lyon, J.D., (1996), Detecting abnormal operating performance: The empirical power and specification of test statistics, *Journal of Financial Economics*, 41, pp 359-399.
- Barney, J., (1991), Firm resources and sustained competitive advantage, *Journal of Management*, 17(1), 99–120.
- Berger, A. N., (2000), The “big picture” of bank diversification. *Proceedings of Federal Reserve Bank of Chicago*, 248, 50–54.
- Bhagwat, V., Dam, R., & Harford, J., (2016), The real effects of uncertainty on merger activity, *Review of Financial Studies*, 29, 3000–3034.
- Biswas, S., & Zhai, W., (2021). Economic policy uncertainty and cross-border lending, *Journal of Corporate Finance*, 67.

- Bonaime, A., Gulen, H., & Ion, M. (2018). Does policy uncertainty affect mergers and acquisitions? *Journal of Financial Economics*, 129, 531–558.
- Boone, A. L., E. Lie, and Y. Liu (2014), Time Trends and Determinants of the Method of Payment in M&As, *Journal of Corporate Finance* 27: 296–304.
- Borthwick, J., Ali, S., & Pan, X. (2020), Does policy uncertainty influence mergers and acquisitions activities in China?, A replication study, *Pacific-Basin Finance Journal*, 62(101381), 1–13.
- Brewer, H. L., (1981), Investor benefits from corporate international diversification, *Journal of Financial and Quantitative Analysis*, 16, 113.
- Brogaard, J., & Detzel, A. (2015), The asset-pricing implications of government economic policy uncertainty, *Management Science*, 61(1), 3–18.
- Cao, C., Li, X., & Liu, G. (2017). Political uncertainty and cross-border acquisitions, *Review of Finance*, 1, 32.
- Da, Z., Engelberg, J., & Gao, P. (2014), The sum of all FEARS investor sentiment and asset prices, *Review of Financial Studies*, 28(1), 1–32.
- Danbolt, J., & Maciver, G., (2012), Cross-Border versus Domestic Acquisitions and the Impact on Shareholder Wealth. *Journal of Business Finance & Accounting*, 39(7-8), 1028–1067.
- Dang, M., Henry, D., Nguyen, M. T., & Hoang, V. A. (2018), Cross-country determinants of ownership choices in cross-border acquisitions: Evidence from emerging markets, *Journal of Multinational Financial Management*, 44(1), 14–35.
- Davis, S. J., (2016), An Index of Global Economic Policy Uncertainty, *Macroeconomic Review*, October.
- Davis, S. J., Dingquian Liu and Xuguang Simon Sheng, (2019), Economic Policy Uncertainty in China Since 1946: The View from Mainland Newspapers, working paper, August.
- Demir, E., & Ersan, O. (2017), Economic policy uncertainty and cash holdings: Evidence from BRIC countries, *Emerging Markets Review*, 33, 189–200.
- Denis, D. J., Denis, D. K., & Yost, K. (2002), Global diversification, industrial diversification, and firm value, *Journal of Finance*, 57, 1951–1979.

- Dessaint, O., Golubov, A., & Volpin, P. (2017), Employment protection and takeovers, *Journal of Financial Economics*, 125, 369–388.
- Dikova, D., Sahib, P. R., & van Witteloostuijn, A. (2010), Cross-border acquisition abandonment and completion: The effect of institutional differences and organizational learning in the international business service industry, 1981–2001, *Journal of International Business Studies*, 41(2): 223-245.
- Dinc, I., & Erel, I. (2013), Economic nationalism in mergers and acquisitions, *The Journal of Finance*, 68(6), 2471–2514.
- Draper, P., & Paudyal, K., (2006), Acquisitions: Private versus Public. *European Financial Management: the Journal of the European Financial Management Association*, 12(1), 57-80.
- Erel, I., Liao, R. C., & Weisbach, M. S. (2012), Determinants of cross-border mergers and acquisitions, *Journal of Finance*, 47, 1045–1082.
- Fatemi, A. M., (1984), Shareholder benefits from corporate international diversification, *Journal of Finance*, 39, 1325.
- Ferreira, M. A., Massa, M., & Matos, P. (2010), Shareholders at the gate? Institutional investors and cross-border mergers and acquisitions, *Review of Financial Studies*, 23, 601–644.
- Fowler, K.L., & Schmidt, D.R., (1988), Tender Offers, Acquisitions, and Subsequent Performance in Manufacturing Firms. *The Academy of Management Journal*, vol. 31, pp. 962-974.
- Fowler, K.L., & Schmidt, D.R., (1989), Determinants of tender offer post-acquisition financial performance, *Strategic Management Journal*, 10, pp 339-350.
- Franks, J. R., Harris, R. S., & Mayer, C. (1988), Means of payment in takeovers: Results for the United Kingdom and the United States, *Corporate takeovers: Causes and consequences*, 221-264.
- Franks, J., Harris, R., Titman, S., (1991), The post merger share-price performance of acquiring firms, *Journal of Financial Economics*, 29, pp 81–96.
- Galbraith, J. K., (1977), *The age of uncertainty*, Boston: Houghton Mifflin.
- Garfinkel, J. A., & Hankins, K. W. (2011), The role of risk management in mergers and merger waves, *Journal of Financial Economics*, 101, 515–532.

- Ghirelli, C., J.J. Perez, and A. Urtasun (2019), A New Economic Policy Uncertainty Index for Spain, Bank of Spain, Working Paper No., 1906.
- Gil, M. and D. Silva (2018), Economic Policy Uncertainty Indices for Colombia, working paper.
- Guardo, D., Chiara, M., Marrocu, E., & Paci, R. (2016), The effect of local corruption on ownership strategy in cross-border mergers and acquisitions, *Journal of Business Research*, 69, 4225–4241.
- Hardouvelis, G. A., G. I. Karalas, D. I. Karanastasis and P. K. Samartzis (2018), Economic Policy Uncertainty, Political Uncertainty and the Greek Economic Crisis, working paper, May.
- Huang, P., Officer, M., & Powell, R. (2016), Method of payment and risk mitigation in cross-border mergers and acquisitions, *Journal of Corporate Finance*, 40, 216–234.
- Huizinga, H. P., & Voget, J. (2009), International taxation and the direction and volume of cross-border m&as, *Journal of Finance*, 64, 1217–1249.
- Hyun, H.-J., & Kim, H. H. (2010), The determinants of Cross-border M&as: The role of institutions and financial development in the gravity model, *The World Economy*, 33, 292–310.
- La Porta, R., L'opez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1998). Law and finance, *Journal of Political Economy*, 106, 1113–1155.
- Le, Q. V., & Zak, P. J. (2006), Political risk and capital flight, *Journal of International Money and Finance*, 25, 308–329.
- Lee, K. H. (2018). Cross-border mergers and acquisitions amid political uncertainty: A bargaining perspective, *Strategic Management Journal*, 39, 2992–3005.
- Lehne, J., Mo, J., & Plekhanov, A., (2014), What determines the quality of economic institutions? Cross-country evidence, European Bank for reconstruction & development.
- Li, X.-M., Zhang, B., & Gao, R. (2015), Economic policy uncertainty shocks and stock–bond correlations: Evidence from the US market, *Economics Letters*, 132, 91–96.
- Linn, Scott C. & Switzer, Jeannette A. (2001), Are cash acquisitions associated with better post combination operating performance than stock acquisitions?, *Journal of Banking & Finance*, Elsevier, vol. 25(6), pages 1113-1138.
- Marshall, B. R., & Anderson, H. D. (2009), Regulation and target takeover returns: Is there a link? *Pacific-Basin Finance Journal*, 17(4), 395–412.

- Morosini, P., Shane, S., & Singh, H. (1998), National Cultural Distance and cross-border acquisition performance, *Journal of International Business Studies*, 29, 137–158.
- Myers, S. & Majluf, N. (1984), Corporate financing and investment decisions when firms have information investors do not have, *Journal of Financial Economics*, 87, pp 355–374.
- Myers, S. C., & Majluf, N.S. (1984), Corporate financing and investment decisions when firms have information that investors do not have, *Journal of Financial Economics*. 13(2), 187-221.
- Nguyen, N., & Phan, H. (2017). Policy uncertainty and mergers and acquisitions, *Journal of Financial and Quantitative Analysis*, 52, 613–644.
- Norbäck, P.-J., & Persson, L. (2008), Globalization and Profitability of Cross-Border Mergers and Acquisitions, *Economic Theory*, 35(2), 241–266.
- Pástor, L., & Veronesi, P. (2012), Uncertainty about government policy and stock prices, *The Journal of Finance*, 67(4), 1219–1264.
- Peng, M. W., Wang, D. Y., & Jiang, Y. (2008), An institution-based view of international business strategy: A focus on emerging economies, *Journal of International Business Studies*, 39(5): 920-936.
- R. Novy-Marx (2014), Predicting anomaly performance with politics, the weather, global warming, sunspots, and the stars, *Journal of Financial Economics*, 112 (2), 137-146.
- Renneboog, L., Szilagyi, P. G., & Vansteenkiste, C. (2017), Creditor rights, claims enforcement, and bond performance in mergers and acquisitions, *Journal of International Business Studies*, 48, 174–194.
- Rossi, S., & Volpin, P. F. (2004), Cross-country determinants of mergers and acquisitions, *Journal of Financial Economics*, 74, 277–304.
- Scotti, C. (2016), Surprise and uncertainty indexes: Real-time aggregation of real-activity macro-surprises, *Journal of Monetary Economics*, 82, 1–19.
- Sha, Y., Kang, C., & Wang, Z. (2020), Economic policy uncertainty and mergers and acquisitions: Evidence from China, *Economic Modelling*, 89, 590–600.
- Shleifer, A., and R.W. Vishny (2003), Stock Market Driven Acquisitions, *Journal of Financial Economics* 70 (3): 295–311.

- Singh, H., & Montgomery, C. A. (1987), Corporate acquisition strategies and economic performance, *Strategic Management Journal*, 8(4), 377-386.
- Sudarsanam, S., (2010), *Creating Value from Mergers and Acquisitions: The Challenges* (2nd ed), Harlow: Prentice Hall.
- Tallman, S. B., (1988), Home country political risk and foreign direct Investment in the United States, *Journal of International Business Studies*, 19, 219–234.
- Uddin, M., & Boateng, A. (2011), Explaining the trends in the UK cross-border mergers & acquisitions: An analysis of macro-economic factors, *International Business Review*, 20, 547–556.
- Wang, Y., Chen, C. R., & Huang, Y. S. (2014), Economic policy uncertainty and corporate investment: Evidence from China. *Pacific-Basin Finance Journal*, 26, 227–243.
- Wansley, J.W., W. R. Lane, and H. C. Yang (1987), Gains to Bidder Firms in Cash and Securities Transactions, *The Financial Review* 22 (4): 403–414.
- Yook, K. C., P. Gangopadhyay, and G. M. McCabe (1999), Information Asymmetry Theory, Management Control Theory, and Method of Payment in Acquisitions, *Journal of Financial Research* 22 (4): 413–427.
- Zalla, R. (2016), *Economic Policy Uncertainty in Ireland*, working paper, 20 September.
- Zhai, W. (2018), *Economic policy uncertainty and cross-border lending*, Working paper: University of Bristol, UK.
- Zhao, X., Ma, H. & Hao, T. (2019), Acquirer size, political connections and mergers and acquisitions performance: Evidence from China, *Studies in Economics and Finance*, Vol. 36 No. 2, pp. 311-332.

8. Appendices

8.1. Variable information table & country sample

Table 8.1.1.: Summary information & sources of the variables utilized in this thesis.

<i>Variable</i>	<i>Information</i>	<i>Source</i>
<i>Dependent variables</i>		
Number of announced deals	LN(1 + number of announced deals per quarter)	Zephyr
Stock_dummy	Equals 1 if full stock payment, 0 otherwise	Zephyr
Stock_proportion	Value of stock payment/Total deal value	Zephyr
<i>Main explanatory variables</i>		
EPU_3months	BBD index - LN[3-month average]	Baker, Bloom & Davis website
EPU_6months	BBD index - LN[6-month average]	Baker, Bloom & Davis website
EPU_12months	BBD index - LN[12-month average]	Baker, Bloom & Davis website
<i>Country-specific control variables</i>		
Quality of institutions	Sum(government effectiveness; regulatory quality; rule of law; control of corruption)	World governance indicators database
EPU * Quality of institutions	EPU * Quality of institutions	
Tax to GDP	Tax income/GDP	World bank database
GDP growth	GDP growth	World bank database
GDP per capita	LN(GDP per capita)	World bank database
Market capitalization to GDP	Ratio of market capitalization over GDP	World bank database
Trade Openness	Sum of imports & exports value to GDP	World bank database
<i>Firm-specific control variables - acquirer</i>		
Firm size	LN (total firm assets)	Zephyr
Market-to-book ratio	Current market value relative to book value	Zephyr
ROA	Net income divided by the firm's total assets	Zephyr
Leverage ratio	Total long-term liabilities/total assets	Zephyr
U.S. SIC codes (Industry)	Standard Industry Classification codes	Zephyr
<i>Deal-specific control variables</i>		
Public target	Equals 1 if target is listed, 0 otherwise	Zephyr
Deal value	LN(Deal value)	Zephyr
Diversifying deal	Equals 1 if the acquirer & target are in the same industry, based on the 2-digit SIC codes, 0 otherwise	Zephyr

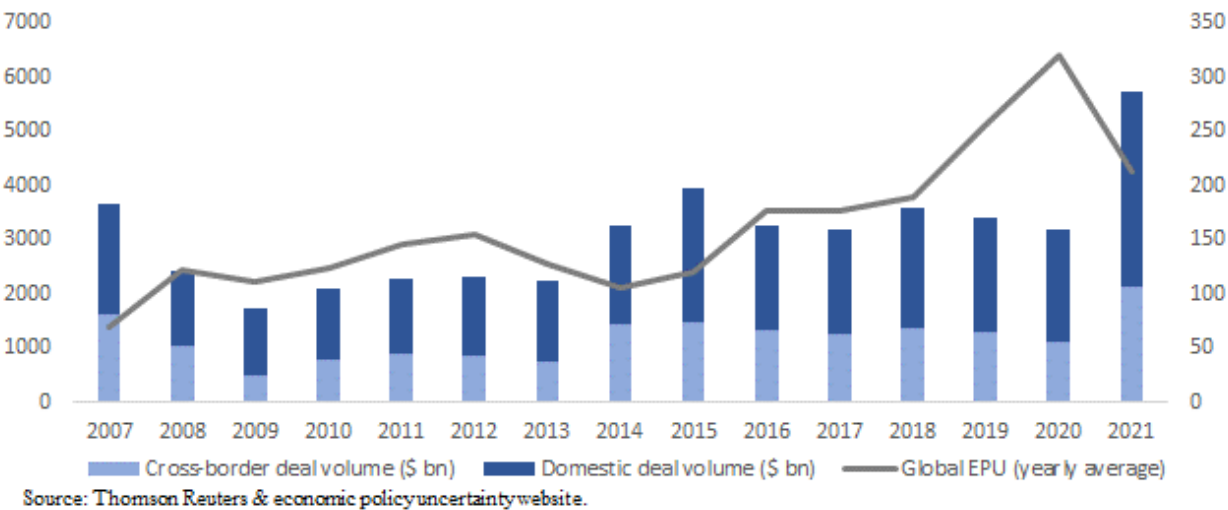
Table 8.1.2.: BBD index Country sample. The sample includes the 24 countries for which the BBD index is available.

Countries for which the BBD index is available (country code)

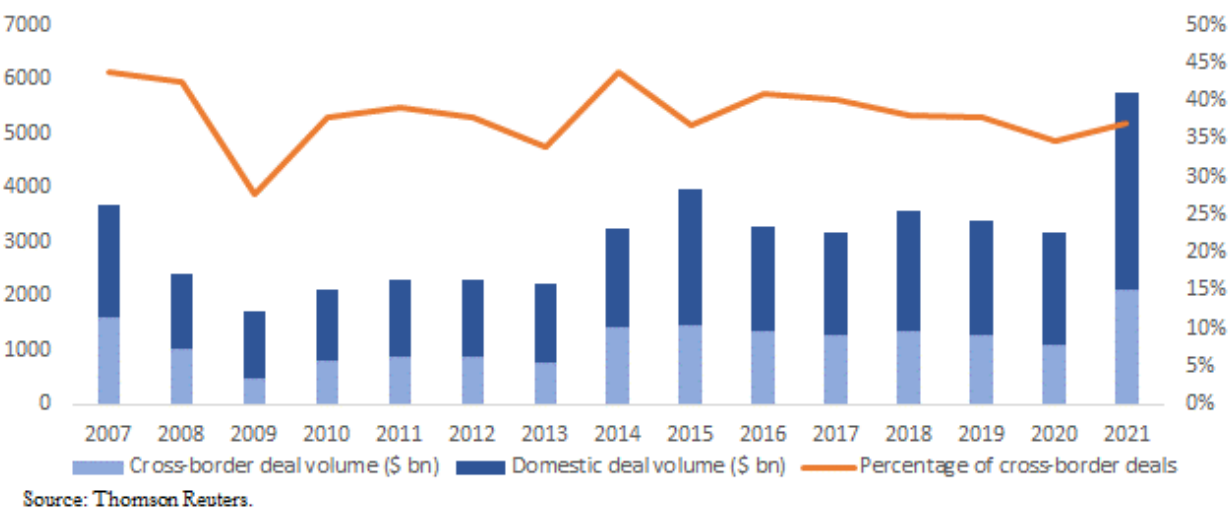
Australia (AU)	Denmark (DK)	Italy (IT)	Spain (ES)
Belgium (BE)	France (FR)	Japan (JP)	United Kingdom (GB)
Brazil (BR)	Germany (DE)	Mexico (MX)	U.S.A.(US)
Canada (CA)	Greece (GR)	Netherlands (NL)	
Chile (CL)	Hong Kong (HK)	Russia (RU)	
China (CN)	India (IN)	Sweden (SE)	
Colombia (CO)	Ireland (IE)	Singapore (SG)	

8.2. M&A & EPU graphs

Graph 8.2.1.: EPU in relation to global M&A deal volume.



Graph 8.2.2.: Domestic & cross-border global M&A activity.



8.3. Pearson correlation tables

Table 8.3.1.: Country-quarter sample - Pearson correlation coefficients. *P*-values are indicated with ***, ** & *, indicating *P*-values smaller than 0.001, 0.01 & the 0.05 respectively.

Variable	Outbound announcements (#)	Inbound announcements (#)	EPU 3months	EPU 6months	EPU 12months	Tax/GDP	GDP growth	GDP per capita	Quality of institutions	Trade openness	Market cap/GDP
Outbound announcements (#)	1										
Inbound announcements (#)	0.791***	1									
EPU_3months	0.339***	0.306***	1								
EPU_6months	0.343***	0.310***	0.967***	1							
EPU_12months	0.349***	0.323***	0.918***	0.964***	1						
Tax/GDP	-0.0924**	-0.264***	-0.00812	-0.00801	-0.00827	1					
GDP growth	0.198***	0.218***	-0.0333	-0.0321	-0.0369	-0.297***	1				
GDP per capita	0.411***	0.171***	0.202***	0.203***	0.205***	0.511***	-0.322***	1			
Quality of institutions	0.498***	0.229***	0.154***	0.153***	0.155***	0.455***	-0.128***	0.844***	1		
Trade openness	0.131***	-0.117***	-0.0311	-0.0339	-0.037	0.119***	0.133***	0.394***	0.466***	1	
Market cap/GDP	0.542***	0.351***	0.101**	0.0977**	0.102**	-0.106**	0.212***	0.297***	0.542***	0.504***	1

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

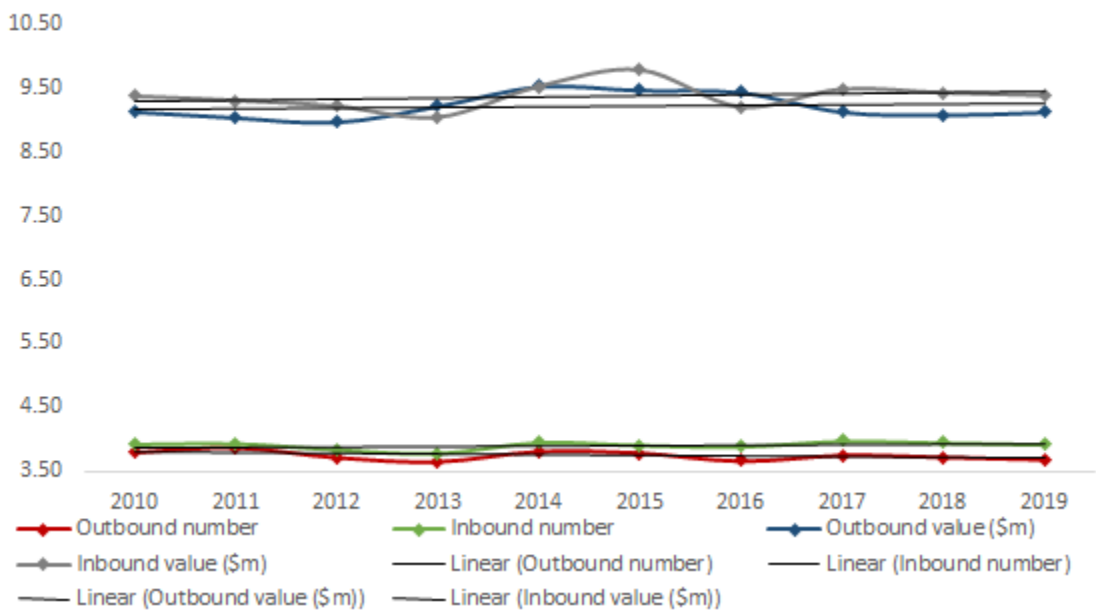
Table 8.3.2.: Cross-border M&A deal sample - Pearson correlation coefficients. *P*-values are indicated with ***, ** & *, indicating *P*-values smaller than 0.001, 0.01 & the 0.05 respectively.

Variable	Stock_dummy	Stock_proportion	EPU 3months	EPU 6months	EPU 12months	Firm size	ROA	Leverage	Market/Book	Tax/GDP	GDP growth	GDP per capita	Quality of institutions	Trade openness	Stock market cap/GDP	Deal value	Public target	Diversifying
Stock_dummy	1																	
Stock_proportion	0.836***	1																
EPU_3months	-0.00352	-0.0029	1															
EPU_6months	0.00157	-0.0071	0.967***	1														
EPU_12months	0.00505	0.0018	0.932***	0.956***	1													
Firm size	-0.371***	-0.499***	0.00763	0.0241	0.043	1												
ROA	-0.408***	-0.503***	0.046	0.0667*	0.0696**	0.606***	1											
Leverage	-0.161***	-0.229***	-0.0105	-0.00951	0.00457	0.497***	0.214***	1										
Market/Book	-0.00375	0.0282	-0.0476	-0.0631*	-0.0613*	0.0146	-0.0701**	0.110***	1									
Tax/GDP	0.0173	0.0331	0.142***	0.159***	0.184***	0.0125	0.00971	0.0447	0.0132	1								
GDP growth	-0.0405	-0.0638*	0.0289	0.0371	-0.0403	0.0171	-0.00643	-0.110***	0.0294	-0.204***	1							
GDP per capita	0.0362	0.0696**	0.125***	0.147***	0.137***	-0.127***	-0.0401	-0.00323	-0.0446	0.226***	-0.514***	1						
Quality of institutions	0.00524	0.0288	0.140***	0.168***	0.165***	-0.0677**	-0.003	0.000196	-0.00103	0.317***	-0.434***	0.885***	1					
Trade openness	-0.0168	-0.0184	-0.0236	-0.0244	-0.0374	-0.0201	0.0396	-0.0276	-0.0276	0.179***	0.0708**	0.125***	0.263***	1				
Stock market cap/GDP	0.0261	0.0493	0.044	0.0302	0.0687**	-0.0598*	-0.0406	-0.047	0.0362	-0.0896***	-0.0425	0.445***	0.528***	0.243***	1			
Deal value	-0.118***	-0.156***	0.0586*	0.0808**	0.0952***	0.637***	0.303***	0.368***	0.0469	0.0725**	-0.130***	0.143***	0.149***	-0.0884***	0.0452	1		
Public target	0.131***	0.104***	0.0426	0.0429	0.0477	0.140***	0.0069	0.0786**	-0.0368	0.0299	-0.0733**	0.0131	0.0707**	-0.0622*	0.0298	0.255***	1	
Diversifying	0.0139	0.0255	-0.0584*	-0.0572*	-0.0591*	-0.0398	0.0402	0.0394	0.0066	0.0115	-0.0304	0.00702	-0.01	-0.0559*	-0.0128	0.0285	0.0710**	1

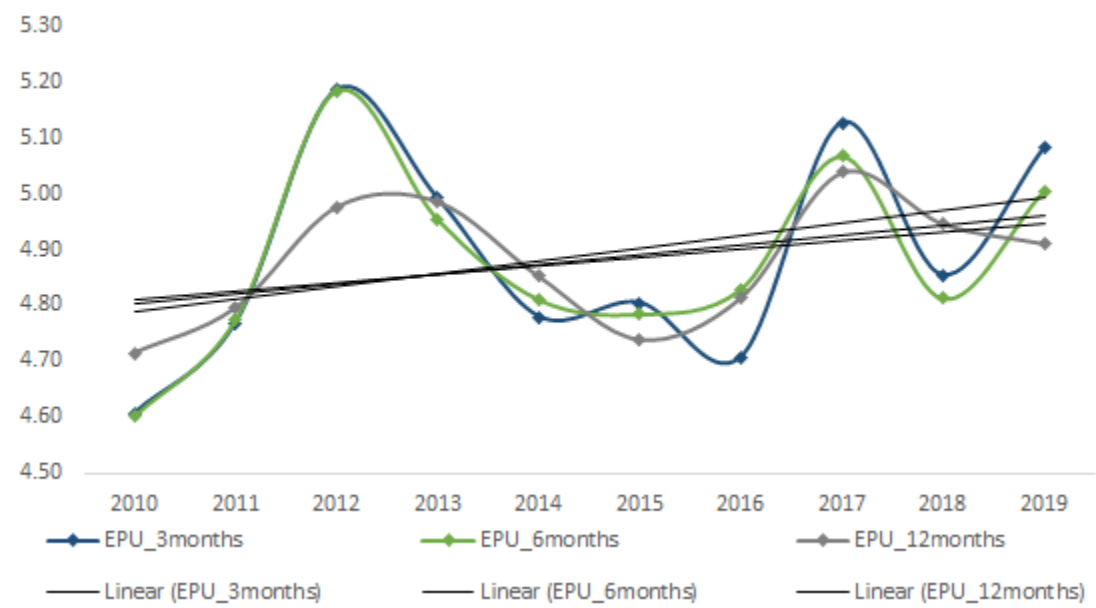
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

8.4. Sample variable graphs

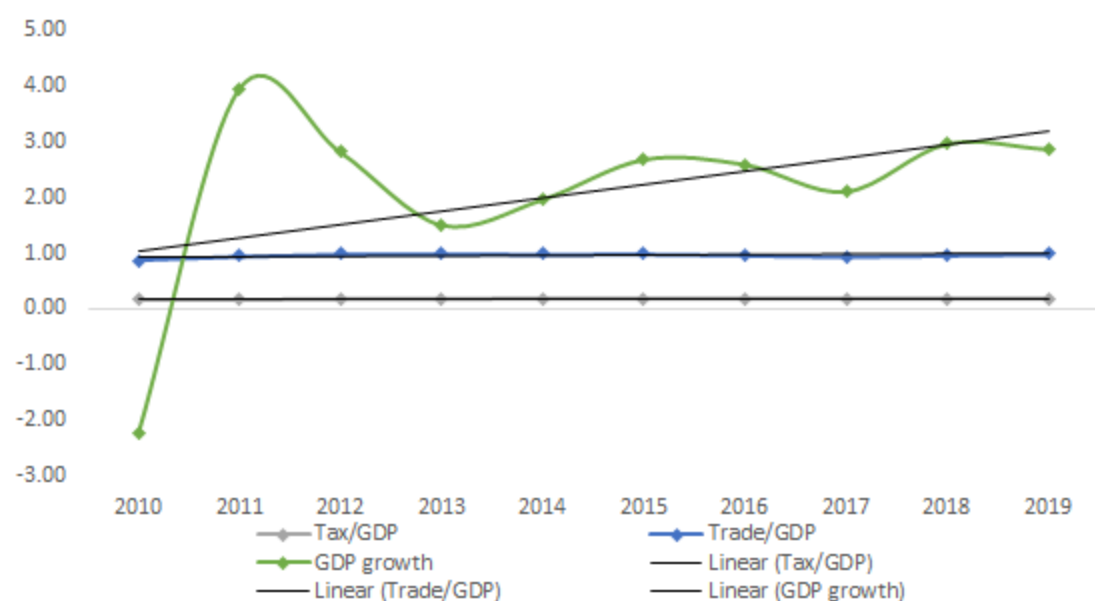
Graph 8.4.1.: Country-year sample | Yearly average variable values [1/4]



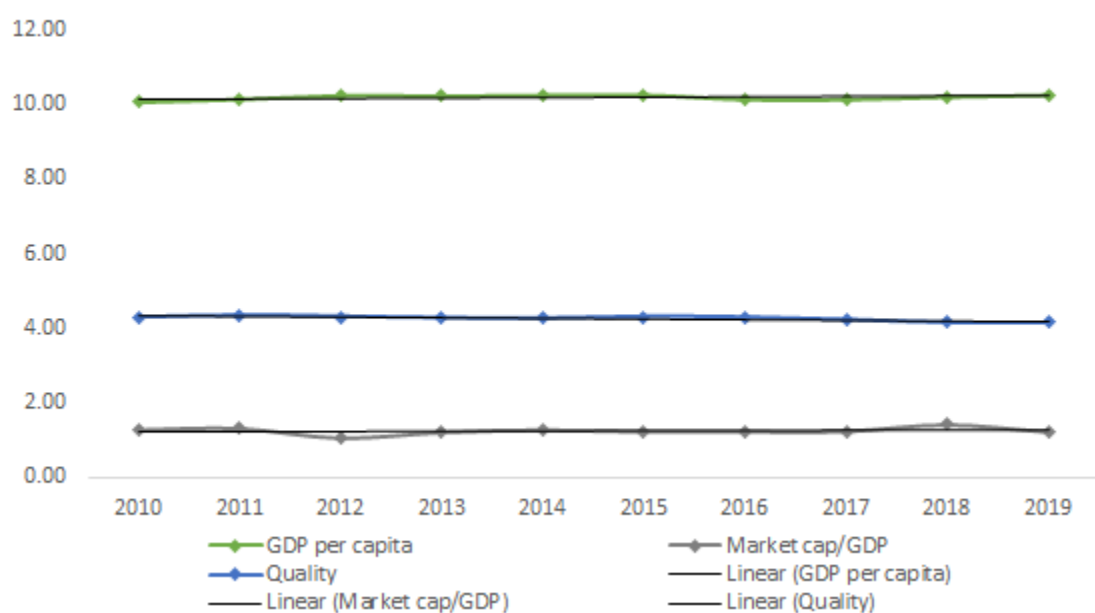
Graph 8.4.1.: Country-year sample | Yearly average variable values [2/4]



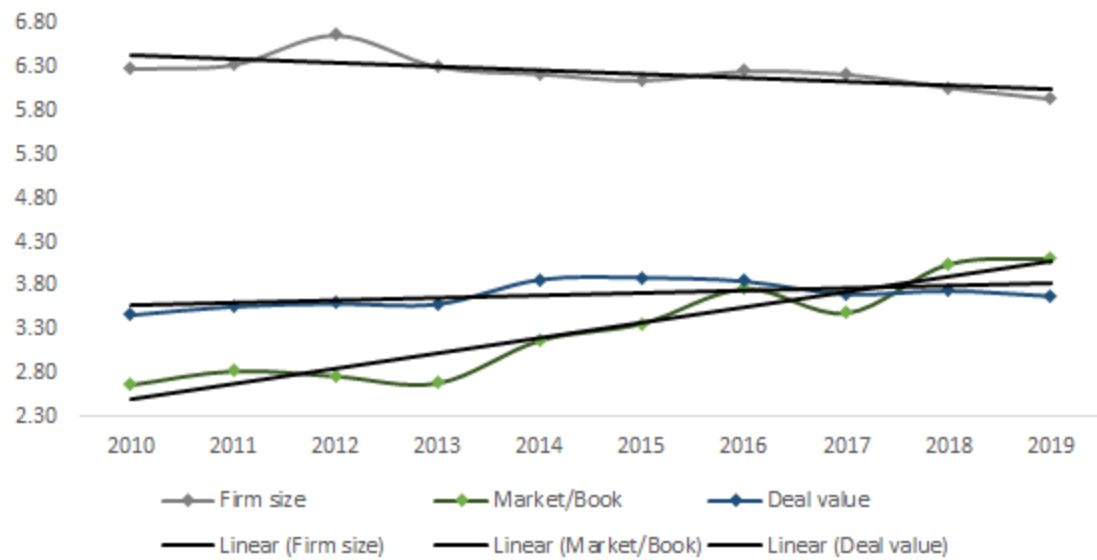
Graph 8.4.1.: Country-year sample | Yearly average variable values [3/4]



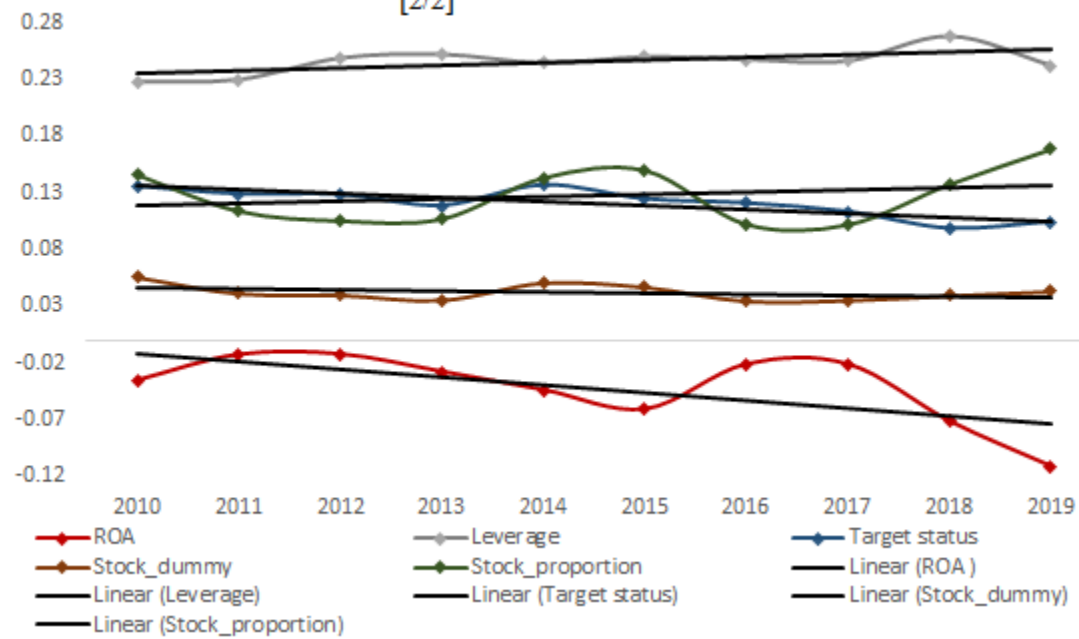
Graph 8.4.1.: Country-year sample | Yearly average variable values [4/4]



Graph 8.4.2.: Cross-border deal sample | Yearly average variable values
[1/2]



Graph 8.4.2.: Cross-border deal sample | Yearly average variable values
[2/2]



8.5. Logit regression coefficients

Table 8.5.1.: Cross-sectional logit regression coefficients. The dependent variable is the *Stock_dummy*, while the z-statistics presented in the parentheses have been calculated based on heteroscedasticity robust, clustered by the acquirer country, standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	Panel A Acquirer country EPU			Panel B Target country EPU		
	(1)	(2)	(3)	(1)	(2)	(3)
EPU_3months	0.2125 (0.5989)			0.7540* (1.8196)		
EPU_6months		0.3544 (0.9658)			1.0115** (2.4346)	
EPU_12months			0.4434 (1.0561)			1.1499*** (3.0378)
<i>Firm characteristics</i>						
Firm size	-0.5120*** (-8.7835)	-0.5093*** (-8.8816)	-0.5099*** (-8.8520)	-0.5331*** (-9.2705)	-0.5391*** (-9.4229)	-0.5435*** (-9.6842)
ROA	-0.8763*** (-5.5140)	-0.8894*** (-5.5761)	-0.8889*** (-5.3652)	-0.3654 (-1.3531)	-0.4010 (-1.3264)	-0.4052 (-1.3253)
Leverage	-1.0631 (-1.3770)	-1.0930 (-1.4241)	-1.0990 (-1.4414)	-0.7270 (-0.7113)	-0.7366 (-0.7569)	-0.7040 (-0.7325)
Market/Book	-0.0278 (-1.0324)	-0.0279 (-1.0222)	-0.0282 (-1.0264)	-0.0249 (-1.0373)	-0.0242 (-1.0684)	-0.0235 (-1.0367)
<i>Country characteristics</i>						
Tax/GDP	-2.1520 (-0.1529)	-1.9791 (-0.1406)	-1.6782 (-0.1183)	0.9486 (0.5982)	0.6005 (0.3835)	0.4015 (0.2815)
Trade/GDP	-3.4394* (-1.7555)	-3.4796* (-1.7881)	-3.4744* (-1.7815)	0.2444 (0.8032)	0.2865 (0.9521)	0.3092 (1.0608)
GDP growth	-0.0512 (-1.5081)	-0.0515 (-1.4596)	-0.0533 (-1.5510)	-0.0568 (-1.1907)	-0.0617 (-1.1805)	-0.0533 (-1.0538)
GDP per capita	-0.7996 (-0.5700)	-0.8501 (-0.6086)	-0.8418 (-0.6058)	0.6303** (2.0131)	0.6821** (2.1666)	0.6933** (2.1520)
Quality of institutions	-0.1393 (-0.3034)	-0.1328 (-0.2936)	-0.1276 (-0.2841)	-0.2620** (-2.5121)	-0.2838*** (-2.7639)	-0.2812*** (-2.8553)
Market cap/GDP	-0.2572 (-0.2385)	-0.2868 (-0.2639)	-0.2906 (-0.2673)	0.4468 (0.9164)	0.4480 (0.9523)	0.3973 (0.8573)
<i>Deal characteristics</i>						
Deal value	0.2552*** (3.5255)	0.2561*** (3.5722)	0.2587*** (3.5594)	0.2469*** (3.8654)	0.2439*** (3.9156)	0.2434*** (3.9232)
Public target	1.4196*** (4.6083)	1.4314*** (4.5710)	1.4379*** (4.6240)	1.4140*** (6.4290)	1.4128*** (6.5362)	1.4038*** (6.3522)
Diversifying	-0.1694 (-0.9061)	-0.1683 (-0.9048)	-0.1691 (-0.9193)	0.4230* (1.7511)	0.4388* (1.6617)	0.4455 (1.5820)
Constant	10.4143 (0.7105)	10.2391 (0.7060)	9.5430 (0.6456)	-9.3411*** (-2.8381)	-10.9679*** (-3.0628)	-11.8281*** (-3.1197)
N	2,261	2,261	2,261	2,288	2,288	2,288
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.372	0.372	0.372	0.376	0.380	0.383

Table 8.5.2.: Cross sectional logit robustness coefficients, excluding the deals that have the U.S. & the U.K. as the host or home country, attributable to table 4.3.2.2., the dependent variable is the *Stock_dummy*. The main explanatory variable in Panel A (Panel B) is the acquirer (target) country EPU. The z-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust standard errors, clustered by the acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

<i>Variables</i>	Panel A <i>Acquirer country EPU</i>			Panel A <i>Target country EPU</i>		
	(1)	(2)	(3)	(1)	(2)	(3)
EPU_3months	0.6437 (0.6308)			0.7215** (2.0207)		
EPU_6months		0.8644 (0.8547)			1.1785*** (3.5391)	
EPU_12months			0.8112 (0.7557)			1.3431*** (5.0253)
<i>Firm characteristics</i>						
Firm size	-0.8950*** (-6.9593)	-0.8994*** (-6.6195)	-0.8996*** (-6.7507)	-0.8391*** (-7.0484)	-0.8728*** (-6.9084)	-0.8827*** (-6.6698)
ROA	0.8077 (1.5999)	0.7794 (1.5494)	0.7937 (1.5783)	0.5986 (1.0739)	0.6997 (1.2372)	0.7247 (1.2762)
Leverage	2.8332*** (2.7257)	2.7638*** (2.6879)	2.7656*** (2.7015)	2.2178** (2.1948)	2.2353*** (2.7855)	2.2735*** (3.0169)
Market/Book	-0.0437 (-1.3375)	-0.0453 (-1.4711)	-0.0448 (-1.4538)	-0.0366 (-0.6743)	-0.0310 (-0.5724)	-0.0267 (-0.5111)
<i>Country characteristics</i>						
Tax/GDP	-18.0825 (-1.0051)	-18.2966 (-0.9912)	-18.5108 (-1.0095)	0.7370 (0.1367)	0.6363 (0.1142)	0.3384 (0.0593)
Trade/GDP	0.8615 (0.2065)	0.7731 (0.1846)	0.7076 (0.1673)	-0.2194 (-0.2864)	-0.1063 (-0.1352)	-0.0749 (-0.0968)
GDP growth	-0.1938*** (-2.7804)	-0.1935*** (-2.8012)	-0.1907*** (-2.8605)	-0.0633* (-1.6728)	-0.0597 (-1.2013)	-0.0473 (-0.8622)
GDP per capita	-0.0860 (-0.0400)	-0.1390 (-0.0662)	-0.0202 (-0.0095)	1.3855** (2.2704)	1.3308* (1.8280)	1.3188* (1.8295)
Quality of institutions	0.8503 (0.6234)	0.8448 (0.6380)	0.8351 (0.6223)	-0.4952*** (-2.6036)	-0.4832** (-2.3056)	-0.4688** (-2.3154)
Market cap/GDP	-0.1007 (-0.0397)	-0.1516 (-0.0598)	-0.0861 (-0.0341)	1.2982 (0.7648)	1.1090 (0.6656)	1.0319 (0.6449)
<i>Deal characteristics</i>						
Deal value	0.2157** (2.0632)	0.2248** (2.1638)	0.2289** (2.1274)	0.2163 (1.0255)	0.2028 (1.0054)	0.1967 (0.9930)
Public target	3.0860*** (5.2167)	3.1230*** (5.0746)	3.0928*** (5.2427)	3.1003*** (3.6294)	3.1366*** (3.5811)	3.1665*** (3.7046)
Diversifying	0.1091 (0.2272)	0.1090 (0.2252)	0.0981 (0.2045)	0.0541 (0.1128)	0.1593 (0.3174)	0.1745 (0.3296)
Constant	-4.2259 (-0.2563)	-4.4327 (-0.2773)	-5.5165 (-0.3303)	-13.5685** (-2.3463)	-15.0169** (-2.2007)	-15.8867** (-2.2659)
N	619	619	619	674	674	674
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.499	0.500	0.499	0.491	0.499	0.502

8.6. Robustness test results

Table 8.6.1.: Country level pooled OLS robustness analysis including the alternative measures of the main explanatory variable, attributable to table 4.3.1.1. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	Panel A				Panel B			
	LN(1+Outbound # of cross-border M&A deals)				LN(1+Inbound # of cross-border M&A deals)			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
EPU_3months	-0.0445 (-0.5197)	-0.0369 (-0.2704)			-0.1336*** (-2.9014)	-0.1750** (-2.2956)		
EPU_12months			-0.0969 (-0.8623)	-0.1353 (-0.7237)			-0.2074** (-2.5582)	-0.2101 (-1.6896)
Tax/GDP	0.5103 (0.2504)	0.5144 (0.2545)	0.3850 (0.1954)	0.3431 (0.1791)	1.5421 (0.6090)	1.5195 (0.6141)	1.3327 (0.5422)	1.3297 (0.5431)
Market cap/GDP	0.5318* (2.0563)	0.5311* (2.0626)	0.5075* (1.9941)	0.5090* (1.9938)	0.1219 (0.9082)	0.1258 (0.9508)	0.0928 (0.7157)	0.0929 (0.7181)
Quality of institutions	-0.0025 (-0.0304)	0.0103 (0.0670)	-0.0078 (-0.0950)	-0.0715 (-0.3709)	-0.0827 (-0.7997)	-0.1518 (-1.1220)	-0.0946 (-0.9673)	-0.0991 (-0.7331)
EPU * Quality of institutions		-0.0026 (-0.1216)		0.0130 (0.4482)		0.0139 (1.0424)		0.0009 (0.0431)
GDP growth	0.0141 (1.7136)	0.0144* (1.8157)	0.0138 (1.6865)	0.0124 (1.5624)	0.0197** (2.3742)	0.0180** (2.1292)	0.0195** (2.4632)	0.0194** (2.3665)
GDP per capita	0.2924 (0.8508)	0.2913 (0.8550)	0.2960 (0.8395)	0.2952 (0.8162)	-0.0522 (-0.2563)	-0.0462 (-0.2279)	-0.0518 (-0.2574)	-0.0519 (-0.2587)
Trade/GDP	-0.3354* (-1.9783)	-0.3428* (-2.0381)	-0.3346* (-1.9824)	-0.3013* (-1.8320)	-0.6213 (-1.4934)	-0.5811 (-1.4051)	-0.6249 (-1.4679)	-0.6225 (-1.4173)
Constant	-0.9898 (-0.2620)	-1.0107 (-0.2661)	-0.7073 (-0.1790)	-0.5338 (-0.1269)	3.8959* (1.7685)	4.0094* (1.8912)	4.3658** (2.0927)	4.3781** (2.2267)
N	880	880	880	880	880	880	880	880
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0409	0.0398	0.0432	0.0430	0.0289	0.0295	0.0359	0.0348

Table 8.6.2.: Country level pooled OLS robustness analysis excluding the U.S. & the U.K. from the country sample, attributable to table 4.3.1.2. Panel A (Panel B) illustrates the regression results attributable to the outbound (inbound) cross-border M&A activity. The dependent variable is the natural logarithm of 1 plus the number of announced deals per country, with a quarterly frequency. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with *, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.**

Variables	Panel A						Panel B					
	LN(1+Outbound # of cross-border M&A deals)						LN(1+Inbound # of cross-border M&A deals)					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
EPU_3months	-0.0021 (-0.0229)	-0.0088 (-0.0686)					-0.1505** (-2.8509)	-0.1822** (-2.3366)				
EPU_6months			-0.0201 (-0.1893)	-0.0401 (-0.2686)					-0.2093*** (-3.0316)	-0.2265** (-2.3274)		
EPU_12months					-0.0517 (-0.4223)	-0.0939 (-0.5344)					-0.2268** (-2.3574)	-0.2241* (-1.7569)
Tax/GDP	1.3809 (0.6933)	1.3810 (0.6939)	1.3422 (0.6787)	1.3403 (0.6816)	1.2487 (0.6542)	1.2259 (0.6528)	1.2491 (0.4726)	1.2497 (0.4817)	1.1270 (0.4390)	1.1253 (0.4426)	0.9774 (0.3830)	0.9788 (0.3833)
Market cap/GDP	0.7306** (2.6165)	0.7327** (2.6319)	0.7168** (2.5285)	0.7238** (2.5515)	0.6981** (2.5230)	0.7087** (2.5519)	0.0474 (0.3319)	0.0578 (0.4014)	0.0001 (0.0006)	0.0061 (0.0435)	0.0098 (0.0717)	0.0091 (0.0663)
Quality of institutions	-0.0104 (-0.1271)	-0.0239 (-0.1613)	-0.0106 (-0.1292)	-0.0513 (-0.3058)	-0.0134 (-0.1613)	-0.1001 (-0.5316)	-0.0731 (-0.6683)	-0.1366 (-0.9586)	-0.0755 (-0.7120)	-0.1105 (-0.8019)	-0.0869 (-0.8353)	-0.0815 (-0.5521)
EPU * Quality of institutions		0.0027 (0.1329)		0.0083 (0.3399)		0.0181 (0.6204)		0.0129 (0.8861)		0.0071 (0.4022)		-0.0011 (-0.0467)
GDP growth	0.0157* (1.9461)	0.0154* (1.8521)	0.0156* (1.9224)	0.0147* (1.7333)	0.0153* (1.8935)	0.0135 (1.6214)	0.0197** (2.2845)	0.0182* (2.0697)	0.0192** (2.3216)	0.0184** (2.1467)	0.0191** (2.3498)	0.0192** (2.2720)
GDP per capita	0.4161 (1.2882)	0.4171 (1.2847)	0.4178 (1.2775)	0.4193 (1.2602)	0.4198 (1.2587)	0.4160 (1.2023)	-0.0939 (-0.4413)	-0.0893 (-0.4195)	-0.0923 (-0.4411)	-0.0911 (-0.4339)	-0.0957 (-0.4556)	-0.0954 (-0.4591)
Trade/GDP	-0.4194** (-2.3787)	-0.4119** (-2.3730)	-0.4155** (-2.3660)	-0.3939** (-2.2659)	-0.4134** (-2.3662)	-0.3685** (-2.1850)	-0.5771 (-1.3803)	-0.5415 (-1.2996)	-0.5623 (-1.3152)	-0.5437 (-1.2488)	-0.5789 (-1.3527)	-0.5816 (-1.3084)
Constant	-2.7777 (-0.7815)	-2.7626 (-0.7679)	-2.6930 (-0.7394)	-2.6326 (-0.7027)	-2.5189 (-0.6727)	-2.3243 (-0.5824)	4.2692* (1.8497)	4.3408* (1.9293)	4.5925* (2.0623)	4.6442** (2.1510)	4.7896** (2.2066)	4.7775** (2.3213)
N	800	800	800	800	800	800	800	800	800	800	800	800
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0591	0.0580	0.0593	0.0585	0.0600	0.0603	0.0266	0.0267	0.0334	0.0325	0.0326	0.0314

Table 8.6.3.: Cross sectional logit (marginal effects) & tobit robustness regressions using different measures of EPU regarding hypothesis number four and five (attributable to table 4.3.2.1.). The dependent variable for specifications (1) & (2) is the Stock_dummy, while for (3) & (4) is the Stock_proportion. The main explanatory variable in Panel A (Panel B) is the acquirer (target) country EPU. The z-statistics (t-statistics) presented in the parentheses for the logit (tobit) model have been calculated based on heteroscedasticity-robust standard errors, clustered by the acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	Panel A				Panel A			
	Acquirer country EPU				Target country EPU			
	Logit marginal effects (1)	(2)	Tobit (3)	(4)	Logit marginal effects (1)	(2)	Tobit (3)	(4)
EPU_3months	0.0089 (0.5913)		0.0349 (1.3448)		0.0312* (1.9566)		0.0302*** (3.0539)	
EPU_12months		0.0185 (1.0172)		0.0518 (1.6258)		0.0471*** (3.4690)		0.0496*** (6.4903)
<i>Firm characteristics</i>								
Firm size	-0.0213*** (-10.9164)	-0.0212*** (-10.5961)	-0.0533*** (-8.6121)	-0.0532*** (-8.3617)	-0.0221*** (-13.5099)	-0.0223*** (-15.1611)	-0.0570*** (-5.4699)	-0.0571*** (-5.5228)
ROA	-0.0365*** (-6.6490)	-0.0370*** (-6.5451)	-0.2938*** (-5.6114)	-0.2939*** (-5.5508)	-0.0151 (-1.4086)	-0.0166 (-1.3871)	-0.1893*** (-2.6587)	-0.1905*** (-2.6633)
Leverage	-0.0443 (-1.2659)	-0.0458 (-1.3102)	-0.0633 (-1.3208)	-0.0649 (-1.3659)	-0.0301 (-0.6954)	-0.0289 (-0.7147)	-0.0169 (-0.2460)	-0.0153 (-0.2233)
Market/Book	-0.0012 (-0.9806)	-0.0012 (-0.9770)	0.0007 (0.2360)	0.0008 (0.2590)	-0.0010 (-1.0009)	-0.0010 (-0.9963)	-0.0001 (-0.0539)	-0.0001 (-0.0439)
<i>Country characteristics</i>								
Tax/GDP	-0.0897 (-0.1532)	-0.0699 (-0.1186)	0.5339 (1.1187)	0.5447 (1.1110)	0.0392 (0.5899)	0.0165 (0.2794)	0.2577* (1.7764)	0.2351 (1.6389)
Trade/GDP	-0.1434* (-1.7299)	-0.1448* (-1.7412)	-0.2541 (-1.3837)	-0.2527 (-1.3800)	0.0101 (0.8207)	0.0127 (1.0951)	0.0172 (1.4491)	0.0193 (1.6385)
GDP growth	-0.0021 (-1.6333)	-0.0022* (-1.7165)	-0.0003 (-0.0755)	-0.0007 (-0.1734)	-0.0024 (-1.2447)	-0.0022 (-1.1063)	-0.0019 (-0.5809)	-0.0019 (-0.5824)
GDP per capita	-0.0333 (-0.5869)	-0.0351 (-0.6229)	-0.1242 (-1.3601)	-0.1268 (-1.3518)	0.0261** (1.9688)	0.0284** (2.0919)	0.0093 (0.4177)	0.0100 (0.4297)
Quality of institutions	-0.0058 (-0.2990)	-0.0053 (-0.2802)	0.0500 (1.5608)	0.0525* (1.6559)	-0.0108** (-2.4635)	-0.0115*** (-2.7685)	-0.0130 (-1.3373)	-0.0134 (-1.3729)
Market cap/GDP	-0.0107 (-0.2394)	-0.0121 (-0.2679)	-0.0357 (-0.5152)	-0.0410 (-0.5796)	0.0185 (0.8820)	0.0163 (0.8243)	0.0355 (1.2313)	0.0339 (1.1932)
<i>Deal characteristics</i>								
Deal value	0.0106*** (2.8752)	0.0108*** (2.8472)	0.0329*** (8.9576)	0.0332*** (9.0916)	0.0102*** (3.6295)	0.0100*** (3.6487)	0.0305*** (8.3747)	0.0303*** (8.4838)
Public target	0.0592*** (4.7735)	0.0599*** (4.9892)	0.0722** (2.2734)	0.0733** (2.3131)	0.0585*** (6.5518)	0.0575*** (6.4514)	0.0803*** (2.6478)	0.0801*** (2.6699)
Diversifying	-0.0071 (-0.9286)	-0.0070 (-0.9383)	-0.0260 (-1.3052)	-0.0263 (-1.3251)	0.0175* (1.6901)	0.0183 (1.5202)	0.0191** (2.1320)	0.0202** (2.2516)
Constant	N/A N/A	N/A N/A	1.4997 (1.2030)	1.4203 (1.0952)	N/A N/A	N/A N/A	0.4772** (2.0921)	0.3781 (1.5094)
N	2,261	2,261	1,448	1,448	2,288	2,288	1,448	1,448
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.372	0.372	0.692	0.692	0.376	0.383	0.696	0.703

Table 8.6.4.: Cross sectional logit (marginal effects) & tobit robustness regressions excluding the deals that have the U.S. & the U.K. as the host or home country, attributable to table 4.3.2.2. The dependent variable for specifications (1), (2) & (3) is the *Stock_dummy*, while for (4), (5) & (6) the *Stock_proportion*. The main explanatory variable in Panel A (Panel B) is the acquirer (target) country EPU. The z-statistics (t-statistics) presented in the parentheses for the logit (tobit) model have been calculated based on heteroscedasticity-robust standard errors, clustered by the acquirer country. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	Panel A Acquirer country EPU						Panel B Target country EPU					
	Logit marginal effects			Tobit			Logit marginal effects			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
EPU_3months	0.0335 (0.6764)			0.0709* (1.6899)			0.0365** (2.3061)			0.0481*** (2.9121)		
EPU_6months		0.0448 (0.9399)			0.0754* (1.7809)			0.0588*** (4.5100)			0.0577*** (3.1020)	
EPU_12months			0.0421 (0.8204)			0.0905* (1.9149)			0.0666*** (7.3741)			0.0702*** (4.5624)
<i>Firm characteristics</i>												
Firm size	-0.0465*** (-6.3660)	-0.0466*** (-6.4895)	-0.0467*** (-6.4227)	-0.0649*** (-3.9021)	-0.0652*** (-3.9107)	-0.0653*** (-3.8819)	-0.0424*** (-8.0746)	-0.0436*** (-7.4900)	-0.0437*** (-7.2358)	-0.0583*** (-3.2628)	-0.0582*** (-3.3018)	-0.0584*** (-3.3489)
ROA	0.0420 (1.5299)	0.0404 (1.4867)	0.0412 (1.5141)	-0.1812** (-1.9804)	-0.1808** (-1.9743)	-0.1803** (-1.9761)	0.0303 (1.0370)	0.0349 (1.1853)	0.0359 (1.2296)	-0.1676* (-1.7277)	-0.1677* (-1.7403)	-0.1662* (-1.7386)
Leverage	0.1473*** (2.5795)	0.1433*** (2.5761)	0.1435** (2.5417)	0.1462*** (2.9049)	0.1461*** (2.8506)	0.1440*** (2.7942)	0.1121** (2.4867)	0.1116*** (3.3347)	0.1127*** (3.6240)	0.1923*** (2.4713)	0.1901** (2.4472)	0.1887** (2.4681)
Market/Book	-0.0023 (-1.2179)	-0.0023 (-1.3337)	-0.0023 (-1.3209)	-0.0026 (-0.5578)	-0.0027 (-0.5684)	-0.0027 (-0.5884)	-0.0018 (-0.6544)	-0.0015 (-0.5577)	-0.0013 (-0.5004)	-0.0027 (-0.7012)	-0.0025 (-0.6338)	-0.0021 (-0.5561)
<i>Country characteristics</i>												
Tax/GDP	-0.9400 (-0.9992)	-0.9489 (-0.9884)	-0.9607 (-1.0087)	-0.0294 (-0.0241)	-0.0446 (-0.0368)	-0.0626 (-0.0509)	0.0373 (0.1378)	0.0318 (0.1149)	0.0168 (0.0595)	0.3281* (1.6865)	0.3329* (1.7770)	0.3377* (1.8076)
Trade/GDP	0.0448 (0.2049)	0.0401 (0.1831)	0.0367 (0.1660)	0.1669 (1.0242)	0.1602 (0.9893)	0.1570 (0.9742)	-0.0111 (-0.2849)	-0.0053 (-0.1348)	-0.0037 (-0.0967)	0.0475* (1.6857)	0.0507* (1.7982)	0.0530** (1.9773)
GDP growth	-0.0101*** (-3.3942)	-0.0100*** (-3.4404)	-0.0099*** (-3.4100)	-0.0049 (-0.6656)	-0.0050 (-0.6747)	-0.0050 (-0.6818)	-0.0032* (-1.8865)	-0.0030 (-1.3087)	-0.0023 (-0.9098)	-0.0024 (-0.4742)	-0.0029 (-0.5696)	-0.0024 (-0.4879)
GDP per capita	-0.0045 (-0.0400)	-0.0072 (-0.0663)	-0.0010 (-0.0095)	-0.1157 (-0.8889)	-0.1059 (-0.8030)	-0.0945 (-0.7018)	0.0700** (1.9827)	0.0664 (1.6414)	0.0653* (1.6541)	0.0409 (0.8210)	0.0384 (0.7768)	0.0378 (0.7834)
Quality of institutions	0.0442 (0.6168)	0.0438 (0.6313)	0.0433 (0.6145)	0.1245* (1.9011)	0.1250* (1.9150)	0.1225* (1.9103)	-0.0250** (-2.2416)	-0.0241** (-2.0208)	-0.0232** (-2.0445)	-0.0227 (-1.4302)	-0.0228 (-1.4584)	-0.0222 (-1.4533)
Market cap/GDP	-0.0052 (-0.0396)	-0.0079 (-0.0597)	-0.0045 (-0.0340)	-0.2145 (-1.3513)	-0.2120 (-1.3493)	-0.2079 (-1.3329)	0.0656 (0.7379)	0.0553 (0.6447)	0.0511 (0.6269)	-0.0014 (-0.0200)	-0.0045 (-0.0641)	-0.0100 (-0.1470)
<i>Deal characteristics</i>												
Deal value	0.0112* (1.8691)	0.0117** (1.9648)	0.0119** (1.9674)	0.0332*** (2.7056)	0.0334*** (2.7342)	0.0340*** (2.7820)	0.0109 (0.9995)	0.0101 (0.9825)	0.0097 (0.9750)	0.0293*** (2.7181)	0.0292*** (2.7450)	0.0287*** (2.7095)
Public target	0.1604*** (7.3251)	0.1620*** (7.5468)	0.1605*** (7.3734)	0.1214*** (3.1489)	0.1226*** (3.1841)	0.1244*** (3.2862)	0.1567*** (4.3773)	0.1565*** (4.3033)	0.1569*** (4.3854)	0.1211*** (3.5210)	0.1210*** (3.5432)	0.1214*** (3.5984)
Diversifying	0.0057 (0.2252)	0.0057 (0.2229)	0.0051 (0.2027)	-0.0168 (-0.5477)	-0.0171 (-0.5628)	-0.0186 (-0.6301)	0.0027 (0.1127)	0.0080 (0.3158)	0.0086 (0.3282)	-0.0330 (-1.0263)	-0.0325 (-0.9993)	-0.0322 (-0.9894)
Constant	N/A	N/A	N/A	0.8871 (0.7086)	0.7628 (0.6087)	0.5722 (0.4458)	N/A	N/A	N/A	-0.1518 (-0.5839)	-0.1773 (-0.6840)	-0.2362 (-0.8926)
N	619	619	619	530	530	530	674	674	674	530	530	530
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R ²	0.499	0.500	0.499	0.839	0.840	0.839	0.491	0.499	0.502	0.831	0.839	0.842

8.7. Quality of institutions variable breakdown (country level analysis)

Table 8.7.1.: Quality of institutions variable breakdown - country level pooled OLS results, regarding the relationship between EPU and the outbound M&A activity. The “Interaction term” represents the product of the EPU variable and the respective Quality of institution constituent variable. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	LN(1+Outbound # of cross-border M&A deals)											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
EPU_6months	-0.0660 (-0.6672)	-0.0751 (-0.4707)	-0.0667 (-0.6932)	-0.1069 (-0.7032)	-0.0615 (-0.6361)	-0.0758 (-0.5567)	-0.0632 (-0.6543)	-0.0739 (-0.4903)	-0.0665 (-0.7082)	-0.0464 (-0.2836)	-0.0654 (-0.6684)	-0.1672 (-0.4390)
Tax/GDP	0.4822 (0.2380)	0.4759 (0.2385)	0.5826 (0.3047)	0.5862 (0.3068)	0.2736 (0.1348)	0.2546 (0.1282)	0.5690 (0.2693)	0.5688 (0.2705)	0.4903 (0.2214)	0.5114 (0.2353)	1.4733 (0.5936)	1.4156 (0.5938)
Market cap/GDP	0.5178* (1.9828)	0.5187* (1.9917)	0.5099* (2.0391)	0.5093* (2.0393)	0.5439** (2.1303)	0.5464** (2.1480)	0.5010* (1.9886)	0.5014* (1.9871)	0.5184* (1.8697)	0.5147* (1.9074)	0.4815* (1.8085)	0.4825* (1.8043)
Quality of institutions	-0.0032 (-0.0385)	-0.0184 (-0.1064)										
Government effectiveness			0.0475 (0.1519)	-0.1710 (-0.2690)								
Control of corruption					-0.1231 (-0.6423)	-0.2352 (-0.5300)						
Rule of law							0.1974 (1.2172)	0.1264 (0.2485)				
Regulatory quality									-0.0141 (-0.0591)	0.1143 (0.2320)		
Investor protection											-0.0942 (-0.9780)	-0.1689 (-0.5850)
Interaction term		0.0031 (0.1242)		0.0468 (0.5632)		0.0221 (0.2951)		0.0158 (0.1600)		-0.0250 (-0.2306)		0.0159 (0.3317)
GDP growth	0.0140 (1.7016)	0.0136 (1.7021)	0.0139 (1.6918)	0.0126 (1.5756)	0.0138 (1.6910)	0.0131 (1.6639)	0.0144* (1.7816)	0.0140* (1.7739)	0.0140 (1.7176)	0.0147* (1.8353)	0.0164* (1.7642)	0.0159* (1.7955)
GDP per capita	0.2946 (0.8489)	0.2954 (0.8479)	0.2835 (0.8860)	0.2807 (0.8616)	0.3168 (0.9477)	0.3179 (0.9387)	0.2559 (0.7687)	0.2562 (0.7626)	0.2965 (0.7782)	0.2914 (0.7903)	0.2664 (0.7687)	0.2751 (0.8113)
Trade/GDP	-0.3313* (-1.9599)	-0.3228* (-1.9174)	-0.3163** (-2.1907)	-0.2802** (-2.1443)	-0.3631** (-2.6870)	-0.3482** (-2.4299)	-0.2613* (-1.7228)	-0.2491 (-1.4919)	-0.3319* (-1.8608)	-0.3482 (-1.7047)	-0.2981** (-2.2891)	-0.2855* (-1.8584)
Constant	-0.8917 (-0.2313)	-0.8610 (-0.2179)	-0.8632 (-0.2297)	-0.6800 (-0.1747)	-0.9877 (-0.2594)	-0.9371 (-0.2388)	-0.7822 (-0.2046)	-0.7499 (-0.1910)	-0.9079 (-0.2227)	-0.9488 (-0.2337)	-0.1970 (-0.0468)	0.1903 (0.0387)
N	880	880	880	880	880	880	880	880	880	880	880	880
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417	0.0417

Table 8.7.2.: Quality of institutions variable breakdown - country level pooled OLS results, regarding the relationship between EPU and the inbound M&A activity. The “Interaction term” represents the product of the EPU variable and the respective Quality of institution constituent variable. The t-statistics presented in the parentheses have been calculated based on heteroscedasticity-robust clustered by country standard errors. Significance is indicated with ***, ** & *, indicating significance at respectively the 0.01, 0.05 & the 0.1 level.

Variables	LN(1+Inbound # of cross-border M&A deals)											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
EPU_6months	-0.1857*** (-3.1367)	-0.2149** (-2.2817)	-0.1821*** (-3.1348)	-0.2351** (-2.4350)	-0.1824*** (-3.1413)	-0.1891** (-2.1144)	-0.1914*** (-3.2976)	-0.2034** (-2.1987)	-0.1828*** (-3.0628)	-0.2282** (-2.5230)	-0.1857*** (-3.1224)	-0.3539 (-1.3246)
Tax/GDP	1.4757 (0.5968)	1.4555 (0.5982)	1.4297 (0.5900)	1.4344 (0.6004)	1.6903 (0.6724)	1.6814 (0.6710)	1.6914 (0.7383)	1.6913 (0.7425)	1.8774 (0.7201)	1.8298 (0.7168)	1.5285 (0.5193)	1.4331 (0.4925)
Market cap/GDP	0.0874 (0.6557)	0.0903 (0.6865)	0.0647 (0.4666)	0.0638 (0.4640)	0.0557 (0.4384)	0.0569 (0.4488)	0.0680 (0.5503)	0.0685 (0.5544)	0.0232 (0.1610)	0.0316 (0.2198)	0.0471 (0.4036)	0.0487 (0.4254)
Quality of institutions	-0.0846 (-0.8449)	-0.1332 (-1.0415)										
Government effectiveness			-0.2298 (-1.0659)	-0.5171 (-1.3341)								
Control of corruption					-0.0874 (-0.3761)	-0.1397 (-0.3168)						
Rule of law							-0.4245 (-1.3871)	-0.5047 (-1.2529)				
Regulatory quality									0.0710 (0.2883)	-0.2183 (-0.5278)		
Investor protection											0.0308 (0.4071)	-0.0928 (-0.5012)
Interaction term		0.0098 (0.6132)		0.0615 (0.9569)		0.0103 (0.1636)		0.0179 (0.2694)		0.0563 (0.9001)		0.0263 (0.7039)
GDP growth	0.0195** (2.4224)	0.0183** (2.2149)	0.0198** (2.4843)	0.0181** (2.2694)	0.0195** (2.4773)	0.0191** (2.3892)	0.0187** (2.2321)	0.0182** (2.1012)	0.0195** (2.4385)	0.0178* (2.0629)	0.0188** (2.5640)	0.0180** (2.3715)
GDP per capita	-0.0476 (-0.2371)	-0.0451 (-0.2242)	-0.0833 (-0.4115)	-0.0869 (-0.4319)	-0.1056 (-0.5276)	-0.1051 (-0.5204)	-0.0464 (-0.2273)	-0.0461 (-0.2263)	-0.1473 (-0.8350)	-0.1358 (-0.7518)	-0.1151 (-0.5697)	-0.1006 (-0.4712)
Trade/GDP	-0.6109 (-1.4380)	-0.5840 (-1.3578)	-0.5664 (-1.3429)	-0.5189 (-1.2095)	-0.5371 (-1.1902)	-0.5302 (-1.1648)	-0.6544* (-1.7504)	-0.6405 (-1.6816)	-0.4899 (-1.1475)	-0.4533 (-1.0475)	-0.5216 (-1.2024)	-0.5009 (-1.1674)
Constant	4.1430* (1.9389)	4.2413* (2.0749)	4.3742* (2.0330)	4.6151** (2.2028)	4.3795* (1.9958)	4.4031* (2.0604)	4.2455* (1.9501)	4.2819* (2.0351)	4.5960** (2.2196)	4.6881** (2.3646)	4.2325* (1.8886)	4.8730** (2.8267)
N	880	880	880	880	880	880	880	880	880	880	880	880
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.0353	0.0349	0.0360	0.0361	0.0331	0.0321	0.0387	0.0378	0.0328	0.0331	0.0329	0.0326